KR2026 Formatting Instructions

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Abstract

We extend the languages of Coalition Logic (CL) and Alternating-time Temporal Logic (ATL) with new modalities for beneficial and harmful strategic capabilities, capturing the potential benefit and potential harm that a coalition can bring to another coalition through its strategic choices. These languages are interpreted over concurrent games enriched with agents' preferences. On the conceptual side, we use these languages to formalize the notions of potential boon and danger. A potential boon corresponds to a coalition having the capability to confer a benefit on another coalition. Danger corresponds to a coalition having the capability to inflict harm on another coalition. On the technical side, we present several results concerning their axiomatization, their relationships with standard CL and ATL, and their computational complexity.

1 Introduction

Understanding how agents can influence each other's outcomes in strategic settings is a central problem in multi-agent systems. In many practical scenarios, agents or coalitions act not only to achieve their own goals but also to affect the well-being of other agents, either positively or negatively. This motivates the study of formal notions that capture both the potential benefits and potential harms that coalitions can generate through their strategic choices.

In this paper, we focus on the concept of danger and its positive counterpart potential boon. While the general theory of danger assumes that dangerousness can be ascribed not only to agents but also to events and situations, in this paper we focus on the dangerousness of a coalition relative to another coalition. Intuitively, a danger corresponds to a coalition that can inflict harm on another coalition by selecting a particular strategy. In this sense the first coalition is a danger for the second coalition. Its positive counterpart potential boon corresponds to a coalition that can confer a benefit on another coalition through its strategic choices. In this sense the first coalition is a potential boon for the second coalition. These notions allow us to reason explicitly about the effects of coalition strategies on others, capturing scenarios in which outcomes are not merely neutral but have qualitative significance for other agents.

Our analysis aligns with two families of philosophical theories concerning harm (and benefit) and danger. On the one hand, we share with the so-called non-comparative causal account of harm (?; ?) the idea that an agent or coalition harms another agent or coalition when the former brings about a bad outcome for the latter. Conversely, it is beneficial when it contributes to securing a good outcome for the latter.[1] On the other hand, we share with the modal non-probabilistic account of danger (?; ?; ?) the idea that an agent or coalition constitutes a danger for another agent or coalition when the former has the potential to harm the latter.[2] In particular, we interpret potential as the existence of a strategy available to a coalition that, if chosen, would bring harm to another coalition.[3]

The notions of good and bad are central to our analysis and, more broadly, to the theory of harm, benefit, danger, and potential boon. In this work, we assume that these two notions can be defined in terms of agents' preferences: an outcome is good (resp. bad) for an agent when it is one that the agent prefers (resp. disvalues).

To formalize this rich conceptual framework, we build on concurrent games enriched with agents' preferences over computations, interpreted as long-term outcomes. We employ this class of structures to define two novel logical languages, based on Coalition Logic (CL) and Alternating-time Temporal Logic (ATL), which enable

[1]The non-comparative account is traditionally contrasted with the comparative counterfactual account, according to which an event or action is harmful to someone insofar as it makes things go worse for them, overall, than they would have been if the event or action had not occurred.

[2]The modal account is traditionally opposed to the probabilistic account, which identifies danger with risk and contrasts it with safety. Thus, while the probabilistic account conceives of danger as a high probability of harm, the modal account focuses instead on mere possibility.

[3]In (?), a counterfactual interpretation of this notion of potential or possibility is proposed. We leave for future work a comparison between our analysis and Williamson's counterfactual theory of danger.

reasoning about the beneficial (resp. harmful) strategic capabilities of coalitions. We refer to these as the CA−ATL and CA−CL languages, denoting extensions of ATL and CL with coalition-affecting capabilities, respectively. They rest on the assumption that a coalition positively (resp. negatively) affects another coalition when the former brings benefit (resp. harm) to the latter through the choice of a particular strategy. Overall, this framework provides a rigorous basis for representing and reasoning about subtle inter-coalitional relationships, capturing the potentially beneficial or harmful influence that one coalition may exert over another.

We present several technical results concerning the axiomatizability and computational complexity of the proposed languages. In particular, the main technical contributions of the paper are threefold. First, we provide an axiomatization of the CA−CL language with respect to the general class of concurrent game models with preferences, together with a notion of beneficial (resp. harmful) strategy, based on the idea that benefiting (resp. harming) a coalition means guaranteeing the best (resp. worst) outcome for it. Second, we present a variant of this axiomatization for the subclass of models in which agents' preferences are assumed to be non-flat (i.e., it is not the case that an agent is indifferent among all outcomes). Third, we establish a polynomial embedding of the CA−ATL language into standard ATL, under the assumption that agents' preferences remain stable over time, as well as polynomial embeddings of the CA−CL language into standard CL, both with and without this assumption. Thanks to these embeddings, we obtain tight complexity results for satisfiability checking.

The paper is organized as follows. Section ?? discusses related work. Section ?? introduces the formal semantics. In Section ??, we use this semantics to formalize the notions of beneficial and harmful strategies. Section ?? presents the CA−ATL and CA−CL languages. Section ?? is devoted to the axiomatization of the CA−CL language. Section ?? establishes the polynomial embeddings of the CA−ATL and CA−CL languages into standard ATL and CL. Finally, Section ?? concludes the paper. Proofs are provided in a technical appendix included in the supplementary material.

2 Related work

On the formal semantics side, our work draws on two main traditions. On the one hand, it builds upon the concurrent game semantics traditionally used to interpret ATL (?; ?), CL (?; ?), strategy logic (SL) (?; ?), and STIT (seeing-to-it-that) logic (?; ?). On the other hand, it follows the tradition of modal logics of preferences, including the pure logic of preferences and their dynamics (?; ?), the interaction between preferences and epistemic states (?; ?), and higher-order preferences (?). However, these works do not address the strategic or temporal dimensions of preferences.

The extension of concurrent games with preferences has recently attracted attention as a means to model agents' rationality (?), to capture the game-theoretic concept of iterated elimination of strictly dominated strategies (IESDS) (?), and to establish connections between ATL strategic reasoning and correlated equilibrium (?). In (?), a new class of models—concurrent games with preferences—was introduced, offering a sophisticated framework for representing agents' preferences over outcomes in concurrent games, as well as their temporal dynamics. We adopt this class of models in the present work. In contrast, (?) and (?) focus on a simpler notion of dichotomous preference induced by an agent's goal, where the goal is expressed either directly at the model level (?) or as a parameter of the modalities capturing the outcomes of the IESDS procedure (?). A quantitative notion of goals, based on a fuzzy interpretation, has recently been incorporated into ATL and SL (?; ?), and has been used to represent agents' utilities in the context of mechanism design (?).

In the present paper, we use concurrent games with preferences introduced in (?) to provide a novel logical investigation of the notions of beneficial and harmful strategies, as well as the related concepts of potential boon and danger, which have not previously been explored in the context of logics for multi-agent systems.

On the conceptual side, the work most closely related to ours is (?), which offers a causal analysis of the concept of harm based on the semantics of structural equation models. As emphasized in the introduction, there is a close connection between the notion of harm and the notion of danger that we investigate in this paper.

3 Semantics

In this section, we present the semantics that will later be used to define the concepts of potential benefit and potential harm, and to interpret the formal languages expressing these concepts. We begin with the notion of a multi-agent transition frame. Next, we introduce concurrent game frames, a specific class of multi-agent transition frames that satisfy a set of semantic constraints. Finally, we extend concurrent game frames by incorporating a notion of preference over computations. The notion of a concurrent game with preferences is based on (?).

3.1 Multi-agent transition frame

Concurrent game models can be defined from a more general class of structures, which we call multi-agent transition frames (MATFs). An MATF is parameterized by the number of agents. Let $\mathbb{AGT} = \{a \in \mathbb{N} \mid 0 \leq a < n\}$ denote the set of all agents.

Definition 1 (MATF). A multi agent transition frame (MATF) with respect to $\mathbb{AGT}$ is a tuple $\mathsf{F} = (\mathbb{S}, \mathbb{ACT}, \mathcal{R})$, where

- $\mathbb{S}$ is a non-empty set of states,
- $\mathbb{ACT}$ is a set of actions,
- $\mathcal{R} : \mathbb{ACT}^{\mathbb{AGT}} \to 2^{\mathbb{S}\times\mathbb{S}}$ assign a transition relation on $\mathbb{S}$ to every action profile, where $\mathbb{ACT}^{\mathbb{AGT}} = \{\delta_{\mathbb{AGT}} : \mathbb{AGT} \to \mathbb{ACT}\}$ is the set of action profiles.

Given an MATF as in Definition ??, for any coalition C, a joint action of C is a function $\delta_C : C \to \mathbb{ACT}$, and the set of all joint actions of C is denoted by $\mathbb{ACT}^C$. Note that the empty coalition has a unique joint action, $\emptyset$ (also denoted $\delta_\emptyset$), and action profiles correspond to the joint actions of the grand coalition.

For any joint action δ_C, define

$$\begin{aligned}\mathcal{R}_{\delta_C} =&\big\{(s,s') \in \mathbb{S}\times\mathbb{S} \mid \text{there exists } \delta_{\mathbb{AGT}} \in \mathbb{ACT}^{\mathbb{AGT}} \\ &\text{such that } \delta_C \subseteq \delta_{\mathbb{AGT}} \text{ and } (s,s') \in \mathcal{R}(\delta_{\mathbb{AGT}})\big\}.\end{aligned}$$

We write $(s,s') \in \mathcal{R}_{\delta_C}$ as $s\mathcal{R}_{\delta_C}s'$, and define

$$\mathcal{R}_{\delta_C}(s) = \big\{s' \in \mathbb{S} \mid s\mathcal{R}_{\delta_C}s'\big\}.$$

It can be verified that

$$\mathcal{R}_\emptyset = \bigcup_{\delta_{\mathbb{AGT}} \in \mathbb{ACT}^{\mathbb{AGT}}} \mathcal{R}_{\delta_{\mathbb{AGT}}}.$$

The following definition introduces the concept of available joint action for a coalition. Intuitively, a joint action is available for a coalition at a given state if the coalition can choose it at that state.

Definition 2 (Available joint action). We say that a joint action δ_C is available for a coalition C at a state s if $\mathcal{R}_{\delta_C}(s) \neq \emptyset$.

Two fundamental concepts in the semantics are those of path and computation, which are formally defined as follows.

Definition 3 (Path and computation). A path in a MATF $\mathsf{F} = (\mathbb{S}, \mathbb{ACT}, \mathcal{R})$ is a sequence $\lambda : \mathbb{D} \to \mathbb{S}$ of states, where $\mathbb{D}$ is a nonempty (might infinite) initial segment of the natural numbers, such that

$$\lambda(k)\mathcal{R}_\emptyset\lambda(k+1) \quad \text{for all } k \in \mathbb{D} \text{ such that } k+1 \in \mathbb{D}.$$

The set of all paths in F is denoted by $Path_\mathsf{F}$. The set of all paths starting at a state $s \in \mathbb{S}$ is

$$Path_{\mathsf{F},s} = \{\lambda \in Path_\mathsf{F} \mid \lambda(0) = s\}.$$

A computation (or full path) in F is a path $\lambda \in Path_\mathsf{F}$ such that there is no path $\lambda' \in Path_\mathsf{F}$ for which λ is a proper prefix. The set of all computations in F is denoted by $Comp_\mathsf{F}$. The set of computations starting at a state $s \in \mathbb{S}$ is

$$Comp_{\mathsf{F},s} = \{\lambda \in Comp_\mathsf{F} \mid \lambda(0) = s\}.$$

A partial path is a path that is not a computation. The set of all partial paths in F is denoted by $PPath_\mathsf{F}$, and the set of partial paths starting at state $s \in \mathbb{S}$ is

$$PPath_{\mathsf{F},s} = \{\lambda \in PPath_\mathsf{F} \mid \lambda(0) = s\}.$$

For convenience, we denote the domain of a function f by $\mathsf{dom}(f)$. Given a path λ, we write λ_k for $\lambda(k)$ for all $k \in \mathsf{dom}(\lambda)$, denote the cardinality of $\mathsf{dom}(\lambda)$ by $\mathsf{len}(\lambda)$, and denote the restriction of λ to $\{k' \in \mathbb{N} \mid k' \leq k\}$ by $\lambda|_k$ for all $k \in \mathsf{dom}(\lambda)$.

For a finite path λ, let $\mathsf{END}(\lambda)$ denote the greatest number in $\mathsf{dom}(\lambda)$, and write $\lambda(\mathsf{END}(\lambda))$ as $\mathsf{last}(\lambda)$.

The following fact highlights an interesting property of partial paths, which is guaranteed by the fact that the last state of a partial path always has a successor.

Fact 1. λ is a partial path iff λ is finite and $\mathsf{last}(\lambda)$ is a non-terminal state, i.e., $\mathcal{R}_\emptyset(\mathsf{last}(\lambda)) \neq \emptyset$.

The following definition introduces the crucial notion of a joint strategy for a coalition in a MATF. We define it as a total function from partial paths to joint actions of the coalition. The reason for considering it as total rather than partial is that, in the light of Fact ??, the last state of a partial path always has a successor which is reachable through a joint action of any coalition.

Definition 4 (Joint strategy). A joint strategy of a coalition C in a MATF $\mathsf{F} = (\mathbb{S}, \mathbb{ACT}, \mathcal{R})$ is a total function $\mathcal{F}_C : PPath_\mathsf{F} \to \mathbb{ACT}^C$ such that, for all $\lambda \in \mathsf{dom}(\mathcal{F}_C)$, $\mathcal{F}_C(\lambda)$ is available at $\mathsf{last}(\lambda)$. The set of all joint strategies of coalition C in F is denoted by Str^C_F.

Given an initial state s and a joint strategy for a coalition C, we can compute the set of computations generated by this strategy starting at s.

Definition 5 (Generated computations). Let $\mathsf{F} = (\mathbb{S}, \mathbb{ACT}, \mathcal{R})$ be a MATF, $s \in \mathbb{S}$, and $\mathcal{F}_C \in Str^C_\mathsf{F}$. A computation λ is said to be generated by $\mathcal{F}_C$ at s if and only if:

- $\lambda(0) = s$,
- for all $k \in \mathsf{dom}(\lambda)$, if $k+1 \in \mathsf{dom}(\lambda)$ then $\lambda(k)\mathcal{R}_{\mathcal{F}_C(\lambda|_k)}\lambda(k+1)$.

The set of all computations generated by $\mathcal{F}_C$ at s is denoted by $\mathcal{O}_\mathsf{F}(s, \mathcal{F}_C)$.

3.2 Concurrent game frame

We define concurrent game frames (CGFs) as a subclass of MATFs. In particular, a CGF is a MATF which satisfies the three properties of never-ending interaction, independence of choices and collective choice determinism.

Definition 6 (CGF). A MATF $\mathsf{F} = (\mathbb{S}, \mathbb{ACT}, \mathcal{R})$ is a concurrent game frame (CGF), if it satisfies the following three properties:

Never-ending interaction: the relation $\mathcal{R}_\emptyset$ over $\mathbb{S}$ is serial.[4]

Independence of choices: for all states $s \in \mathbb{S}$, for all coalitions C, C' such that $C \cap C' = \emptyset$, and for all joint actions $\delta_C \in \mathbb{ACT}^C$, $\delta_{C'} \in \mathbb{ACT}^{C'}$, if both δ_C and $\delta_{C'}$ are available at s, then their union $\delta_C \cup \delta_{C'}$ is also available at s.

Collective choice determinism: for all action profile $\delta_{\mathbb{AGT}}$, the relation $\mathcal{R}_{\delta_{\mathbb{AGT}}}$ over $\mathbb{S}$ is deterministic.[5]

Never-ending interaction captures the fact that every state in a CGF has at least one successor. According to independence of choices, agents can never be deprived of available actions due to the choices made by other agents. Finally, collective choice determinism ensures that the outcome of a joint action of all agents is

[4] This means that $\forall s \in \mathbb{S}, \exists s' \in \mathbb{S}$ such that $s\mathcal{R}_\emptyset s'$.

[5] This means that $\forall s, s', s'' \in \mathbb{S}$, if $s\mathcal{R}_{\delta_{\mathbb{AGT}}}s'$ and $s\mathcal{R}_{\delta_{\mathbb{AGT}}}s''$ then $s' = s''$.

uniquely determined. In other words, once all agents have made their choices the next state is uniquely determined.

In a CGF, due to the property of collective choice determinism, the set of computations generated by a strategy for the grand coalition $\mathbb{AGT}$ at a given state is a singleton. The unique computation generated by a strategy $\mathcal{F}_{\mathbb{AGT}}$ of the grand coalition $\mathbb{AGT}$ at state s is denoted by $\wr_{\mathsf{F}}(s, \mathcal{F}_{\mathbb{AGT}})$. Moreover, due to independence of choices, the union of the strategies of two disjoint coalitions is itself a strategy for the union of the two coalitions.

3.3 Concurrent game frame with preferences

We now extend a CGF with a notion of preference over computations, resulting in a structure called a CGF with preferences (CGFP). Specifically, we extend a CGF by introducing, for each agent and for each state, a preference ordering that captures the agent's preference over the computations starting from that state. This notion of preference over computations can be understood as a preference over long-term outcomes, since in a CGF the long-term outcomes available to an agent from a given state coincide with the set of computations originating from that state.

Definition 7 (CGFP). A CGF with preferences (CGFP) is a tuple $\mathsf{P} = (\mathsf{F}, \preceq)$, where:

- F is a CGF, and
- $\preceq$: $\mathbb{AGT} \times \mathbb{S} \to 2^{Comp_{\mathsf{F}} \times Comp_{\mathsf{F}}}$ assigns to every pair of agent i and state s a total preorder over $Comp_{\mathsf{F},s}$.

For any agent i, state s, and computations λ, λ', we write $\lambda \preceq_{i,s} \lambda'$ for $(\lambda, \lambda') \in \preceq(i,s)$. As usual, we write $\lambda \prec_{i,s} \lambda'$ if $\lambda \preceq_{i,s} \lambda'$ and $\lambda' \not\preceq_{i,s} \lambda$. We respectively denote by $\succeq_{i,s}$ and $\succ_{i,s}$ the inverses of $\preceq_{i,s}$ and $\prec_{i,s}$. Respectively, we say that the CGFP $(\mathsf{F}, \preceq)$ has stable, non-flat, well-founded, or converse-well-founded preferences if the following corresponding conditions hold: for all agents $i \in \mathbb{AGT}$:

$$(\mathbf{SP}) \forall s, s' \in \mathbb{S}, \forall \lambda, \lambda' \in Comp_{\mathsf{F},s'} \text{ if } s \mathcal{R}_{\emptyset} s',$$
$$\text{then } \big(\lambda' \preceq_{i,s'} \lambda \text{ iff } s\lambda' \preceq_{i,s} s\lambda\big).$$
$$(\mathbf{NFP})\ \forall s \in \mathbb{S}, \exists \lambda, \lambda' \in Comp_{\mathsf{F},s} \text{ such that } \lambda \prec_{i,s} \lambda'.$$
$$(\mathbf{WFP})\ \forall s \in \mathbb{S}, \exists \lambda \in Comp_{\mathsf{F},s} \text{ such that}$$
$$\forall \in Comp_{\mathsf{F},s} : \lambda \preceq_{i,s} \lambda'.$$
$$(\mathbf{CWFP})\ \forall s \in \mathbb{S}, \exists \lambda \in Comp_{\mathsf{F},s} \text{ such that}$$
$$\forall \in Comp_{\mathsf{F},s} : \lambda \succeq_{i,s} \lambda'.$$

Property **SP**, expressing stable preferences, captures the idea that an agent's preference remains consistent over time: an agent prefers a computation λ to a computation λ' starting at the same state s' if and only if it prefers the precursor of λ (i.e., $s\lambda$) to the precursor of λ' (i.e., $s\lambda'$) at every predecessor s of s'. Stability of preferences is in line with the notion of time consistency of preferences studied in economics, as opposed to time inconsistency (?). Property **NFP**, expressing non-flat preferences, captures the idea that agents are not indifferent among all outcomes. Property **WFP** and **CWFP**, expressing well-founded preferences and converse-well-founded preferences, captures the idea that agents' preferences exist the worst and the best expects.

3.4 Concurrent game models with preferences

We conclude this section by moving from frames to models. In particular, we assume a set a countably infite set of atomic propositions $\mathbb{P}$ denoting atomic facts and supplement a CGFP with a valuation function specifying for every state in the CGFP which propositions are true there.

Definition 8 (CGMP). A concurrent game model with preferences (CGMP) with respect to $\mathbb{AGT}$ and $\mathbb{P}$ is a tuple $\mathsf{M} = (\mathsf{P}, \mathcal{V})$, where

- P is CGFP, and
- $\mathcal{V} : \mathbb{S} \to 2^{\mathbb{P}}$ is a valuation function.

In Definition ??, we introduced two properties for CGFPs: stability (**SP**) and non-flatness (**NFP**) of preferences. These properties are inherited by CGMPs. For $X \subseteq \{\mathbf{SP}, \mathbf{NFP}, \mathbf{WFP}, \mathbf{CWFP}\}$, we denote by $\mathbf{CP}^X$ the class of CGMPs satisfying all properties in X. For example, $\mathbf{CP}^{\{\mathbf{NFP}\}}$ denotes the class of CGMPs with non-flat preferences, while $\mathbf{CP}^{\emptyset}$ denotes the class of all CGMPs.

We illustrate the notion of CGMP with the help of a concrete example.

Example 1. We consider a scenario involving three agents, robots 0, 1, and 2, operating in a small environment composed of three rooms: a left room, a middle room, and a right room. Each of the left and right rooms contains a charging station. Robot 1 is positioned in front of the exit of the right room, guarding it to prevent entry, while robot 2 is stationed in front of the exit of the left room, performing the same role. Initially, robot 0 is located in the middle room (proposition inM_0). During the course of the game, it can move and change its position, entering either the right room (proposition inR_0) or the left room (proposition inL_0). The initial configuration is shown in Figure ??.

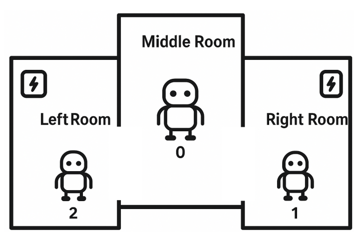
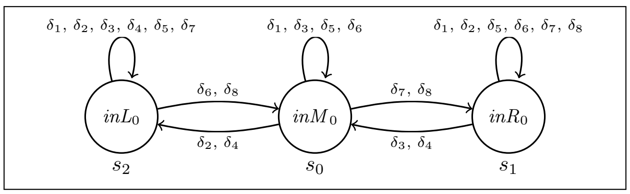


Figure 1: Scenario

Robot 0 can move from its current room either to the right (action ri) or to the left (action le). Thus, to move from the middle room to the right room, it must perform the action ri; conversely, to move from the right room to the middle room, it must perform the action le, and so on. If robot 0 is already in the right (resp. left) room and attempts to move further right (resp. left), it remains in place, as it encounters a wall.

Robots 1 and 2 can either block (action bl) or free (action fr) the passage of the room they are guarding. Thus, robot 1 can block or free the passage of the right room, and robot 2 can do the same for the left room. If robot 1 (resp. robot 2) blocks the passage of its respective room, then robot 0 cannot reach the corresponding charging station when it is in the middle room, and cannot leave the room whose passage is blocked when it is located there.

Robot 0's goal is to reach a charging station at some point in the future, while the goal of robots 1 and 2 is to prevent it from doing so. The CGMP corresponding to this scenario is shown in Figure ??.



(a) State transition diagram.

$$\begin{array}{ll} \delta_1 = (0{:}le, 1{:}bl, 2{:}bl) & \delta_5 = (0{:}ri, 1{:}bl, 2{:}bl) \\ \delta_2 = (0{:}le, 1{:}bl, 2{:}fr) & \delta_6 = (0{:}ri, 1{:}bl, 2{:}fr) \\ \delta_3 = (0{:}le, 1{:}fr, 2{:}bl) & \delta_7 = (0{:}ri, 1{:}fr, 2{:}bl) \\ \delta_4 = (0{:}le, 1{:}fr, 2{:}fr) & \delta_8 = (0{:}ri, 1{:}fr, 2{:}fr) \end{array}$$

(b) Action profiles δ_1–δ_8.

$$\forall s \in \{s_0, s_1, s_2\}, \forall \lambda, \lambda' \in Comp_{\mathsf{F}_{sce}, s}, \forall i \in \{0,1,2\},$$
$$\lambda \preceq_{i,s} \lambda \text{ iff } U_0(\lambda') \leq U_0(\lambda) \text{ with}$$

$$U_0(\lambda) = \begin{cases} 0 & \text{if } \forall k \geq 0, \{inR_0, inL_0\} \cap \mathcal{V}(\lambda(k)) = \emptyset, \\ 1 & \text{otherwise;} \end{cases}$$

$$U_1(\lambda) = U_2(\lambda) = \begin{cases} 0 & \text{if } \exists k \geq 0 \text{ such that} \\ & \{inR_0, inL_0\} \cap \mathcal{V}(\lambda(k)) \neq \emptyset, \\ 1 & \text{otherwise.} \end{cases}$$

(c) Utility-induced preference structure.

Figure 2: CGMP for the scenario: (a) state transitions, (b) action profiles, (c) preference structure. We denote with F_{sce} the underlying CGF.

It consists of three states, s_0, s_1, and s_2, with s_0 as the initial state, and eight possible action profiles, $\delta_1, \ldots, \delta_8$. To make the example self-contained, we assume that each robot's preferences over computations starting from a given state are induced by a dichotomous utility function: this function assigns a value of 1 to a computation that allows the robot to achieve its goal, and a value of 0 otherwise.

4 Beneficial and harmful strategy

In this section, we introduce the concepts of beneficial and harmful strategies. The intuition is straightforward: the strategy of a coalition is said to be beneficial (resp. harmful) for another coalition if the choice of that strategy by the first coalition guarantees a good (resp. bad) outcome for the second coalition. We further distinguish between strongly and weakly beneficial (or harmful) strategies, based on two different interpretations of what constitutes a good or bad outcome for a coalition. A strategy is strongly beneficial (resp. strongly harmful) for a coalition if it guarantees the best (resp. worst) outcome that the agents in the coalition could obtain. Conversely, a strategy is weakly beneficial (resp. weakly harmful) for a coalition if it guarantees an outcome that is better (resp. worse) for the agents in the coalition than the one they could obtain if the strategy were not played.

The following definition introduces the concepts of strongly beneficial and harmful strategy.

Definition 9 (Strongly beneficial and harmful strategy). Given a CGFP $\mathsf{P} = (\mathsf{F}, \preceq)$. We say a strategy $\mathcal{F}_C$ of coalition C is strongly beneficial for coalition $B \subseteq \mathbb{AGT}$ at state s iff $\mathcal{O}_{\mathsf{F}}(s, \mathcal{F}_C) \subseteq Best_{i,s}$ for every $i \in B$, where $Best_{i,s} = \{\lambda \in Comp_{\mathsf{F},s} \mid \forall \lambda' \in Comp_{\mathsf{F},s}(\lambda \not\prec_{i,s} \lambda')\}$.

Similarly, we say a strategy $\mathcal{F}_C$ of coalition C is strongly harmful for coalition $H \subseteq \mathbb{AGT}$ at state s iff $\mathcal{O}_{\mathsf{F}}(s, \mathcal{F}_C) \subseteq Worst_{i,s}$ for every $i \in H$, where $Worst_{i,s} = \{\lambda \in Comp_{\mathsf{F},s} \mid \forall \lambda' \in Comp_{\mathsf{F},s}(\lambda \not\succ_{i,s} \lambda')\}$.

We denote with $StrSB^C_{\mathsf{F}}(s, B)$ and $StrSH^C_{\mathsf{F}}(s, H)$ the set of strongly beneficial strategies of coalition C for coalition B at state s and the set of strongly harmful strategies of coalition C for coalition H at state s, respectively.

Let us now turn to the notion of weakly beneficial and harmful strategies.

Definition 10 (Weakly beneficial and harmful strategy). Given a CGFP $\mathsf{P} = (\mathsf{F}, \preceq)$. We say a strategy $\mathcal{F}_C$ of coalition C is weakly beneficial for coalition $B \subseteq \mathbb{AGT}$ at state s iff $\lambda' \not\succ_{i,s} \lambda$ for all $\lambda \in \mathcal{O}_{\mathsf{F}}(s, \mathcal{F}_C)$, $\lambda' \in Comp_{\mathsf{F},s} - \mathcal{O}_{\mathsf{F}}(s, \mathcal{F}_C)$ and $i \in B$.

Similarly, we say a strategy $\mathcal{F}_C$ of coalition C is weakly harmful for coalition $H \subseteq \mathbb{AGT}$ at state s iff $\lambda' \not\prec_{i,s} \lambda$ for all $\lambda \in \mathcal{O}_{\mathsf{F}}(s, \mathcal{F}_C)$, $\lambda' \in Comp_{\mathsf{F},s} - \mathcal{O}_{\mathsf{F}}(s, \mathcal{F}_C)$ and $i \in H$.

We denote with $StrWB^C_{\mathsf{F}}(s, B)$ and $StrWH^C_{\mathsf{F}}(s, H)$ the set of weakly beneficial strategies of coalition C for coalition B at state s and the set of weakly harmful strategies of coalition C for coalition H at state s, respectively.

The intuition behind Definition ?? is that, for a strategy to be weakly beneficial (resp. harmful) for a coalition, the outcome of the strategy should be better (resp. worse) for each member of the coalition than any outcome they could obtain if a different strategy were played.

Let us return to Example ?? presented in Section ?? in order to illustrate the preceding definitions.

Example 2 (Cont.). Consider the following strategy $\mathcal{F}_{\{1,2\}}$ for the coalition formed by robots 1 and 2:

$$\mathcal{F}_{\{1,2\}}(\lambda) = (1{:}bl, 2{:}bl) \text{ for all } \lambda \in PPath_{\mathsf{F}_{sce}}.$$

This strategy corresponds to blocking the passage of both the right and left rooms indefinitely. It is straightforward to verify that $\mathcal{F}_{\{1,2\}} \in StrSH^{\{1,2\}}_{\mathsf{F}_{sce}}(s_0, \{0\})$, and moreover, $\mathcal{F}_{\{1,2\}} \in StrSB^{\{1,2\}}_{\mathsf{F}_{sce}}(s_0, \{1\})$ and $\mathcal{F}_{\{1,2\}} \in StrSB^{\{1,2\}}_{\mathsf{F}_{sce}}(s_0, \{2\})$. In words, the strategy $\mathcal{F}_{\{1,2\}}$ of coalition $\{1, 2\}$ is strongly harmful for robot 0 at state s_0, and strongly beneficial for both robots 1 and 2 at the same state.

Now consider the following strategy $\mathcal{F}'_{\{1\}}$ for robot 1:

$$\mathcal{F}'_{\{1\}}(\lambda) = (1{:}bl) \text{ for all } \lambda \in PPath_{\mathsf{F}_{sce}}.$$

This strategy corresponds to robot 1 blocking the passage of the right room indefinitely. It is straightforward to verify that $\mathcal{F}'_{\{1\}} \in StrWH^{\{1\}}_{\mathsf{F}_{sce}}(s_0, \{0\})$, and moreover, $\mathcal{F}'_{\{1\}} \notin StrSH^{\{1\}}_{\mathsf{F}_{sce}}(s_0, \{0\})$. In words, the strategy $\mathcal{F}'_{\{1\}}$ of robot 1 is weakly harmful for robot 0 at state s_0, but not strongly harmful.

The following proposition highlights the connection between strongly beneficial/harmful and weakly beneficial/harmful. The proof is in Appendix E of the supplementary material.

Proposition 1. If a strategy $\mathcal{F}_C$ of a coalition C is strongly beneficial (resp. harmful) for a coalition at a state, then it is also weakly beneficial (resp. harmful).

5 Languages

In this section, we leverage the semantics presented above to interpret a novel language that extends the language of ATL with a new family of coalition-affecting strategic capability modalities. These modalities allow us to reason about the strategic choices of one coalition that affect another coalition, under the assumption that the strategy of one coalition affects another insofar as the first coalition's strategic choice has either a positive or a negative effect on the well-being of the second coalition, by bringing it benefit or causing it harm.

The ATL language with coalition-affecting capability modalities, denoted by $\mathcal{L}_{\mathsf{CA-ATL}}(\mathbb{AGT}, \mathbb{P})$, is defined by the following grammar:

$$\begin{array}{lll} \text{State formula:} & \varphi, \psi & ::= p \mid \neg\varphi \mid \varphi \wedge \psi \mid \langle\langle C{:}B{,}H\rangle\rangle\gamma \\ \text{Path formula:} & \gamma & ::= \mathsf{X}\varphi \mid \mathsf{G}\varphi \mid (\varphi \ \mathsf{U} \ \psi) \end{array}$$

where p ranges over $\mathbb{P}$, and C, B and H range over $2^{\mathbb{AGT}}$. The other Boolean connectives and constructs $\vee, \rightarrow, \leftrightarrow, \top, \bot$ are defined as abbreviations in the usual way. For simplicity, we sometimes write $\mathcal{L}_{\mathsf{CA-ATL}}$ instead of $\mathcal{L}_{\mathsf{CA-ATL}}(\mathbb{AGT}, \mathbb{P})$.

Formulas $\langle\langle C{:}B{,}H\rangle\rangle\mathsf{X}\varphi$, $\langle\langle C{:}B{,}H\rangle\rangle\mathsf{G}\varphi$, and $\langle\langle C{:}B{,}H\rangle\rangle(\varphi \ \mathsf{U} \ \psi)$ capture the notion of a coalition-affecting strategic capability to guarantee a certain fact, where this fact is expressed by a temporal formula of LTL (Linear Temporal Logic). In particular, $\langle\langle C{:}B{,}H\rangle\rangle\mathsf{X}\varphi$ has to be read "coalition C has a strategy, which is beneficial for group B, harmful to group H, and guarantees that φ is going to be true in the next state"; $\langle\langle C{:}B{,}H\rangle\rangle\mathsf{G}\varphi$ has to be read "coalition C has a strategy, which is beneficial for group B, harmful to group H, and guarantees that φ will always be true"; $\langle\langle C{:}B{,}H\rangle\rangle(\varphi \ \mathsf{U} \ \psi)$ has to be read "coalition C has a strategy, which is beneficial for group B, harmful to group H, and guarantees that φ will be true until ψ is true".

We can now define the semantic interpretation of formulas in $\mathcal{L}_{\mathsf{CA-ATL}}(\mathbb{AGT}, \mathbb{P})$. As in ATL, formulas are interpreted with respect to pointed models. In our setting, a pointed model is a pair consisting of a CGMP and one of its states. In Section ??, we introduced two notions of beneficial and harmful strategies: strong and weak. Accordingly, two distinct semantic interpretations can be defined—one in which existential quantification ranges over strongly beneficial or harmful strategies, and another in which it ranges over weakly beneficial or harmful ones. The first interpretation is based on what we call the strong satisfaction relation, denoted by $\models$, between state formulas of $\mathcal{L}_{\mathsf{CA-ATL}}(\mathbb{AGT}, \mathbb{P})$ and pairs (M, s) called pointed CGMP, and between path formula of $\mathcal{L}_{\mathsf{CA-ATL}}(\mathbb{AGT}, \mathbb{P})$ and pairs (M, λ) called path-based CGMP, where $\mathsf{M} = (\mathbb{S}, \mathbb{ACT}, \mathcal{R}, \preceq, \mathcal{V})$ is a CGMP, $s \in \mathbb{S}$ and λ is a path in M. (The cases for atomic propositions and Boolean connectives are omitted, as they are defined in the standard way.)

$$\begin{array}{rcl}
(\mathsf{M}, s) \models \langle\langle C{:}B{,}H\rangle\rangle\gamma & \Leftrightarrow & \exists \mathcal{F}_C \in StrSB^C_{\mathsf{M}}(s, B) \\
& & \cap StrSH^C_{\mathsf{M}}(s, H) \text{ s.t. } \\
& & \forall \lambda \in \mathcal{O}(s, \mathcal{F}_C), \big(\mathsf{M}, \lambda\big) \models \gamma; \\
(\mathsf{M}, \lambda) \models \mathsf{X}\varphi & \Leftrightarrow & \big(\mathsf{M}, \lambda(1)\big) \models \varphi, \\
(\mathsf{M}, \lambda) \models \mathsf{G}\varphi & \Leftrightarrow & \forall k \in \mathsf{dom}(\lambda), \text{ if } k > 0 \\
& & \text{then } \big(\mathsf{M}, \lambda(k)\big) \models \varphi, \\
(\mathsf{M}, \lambda) \models (\varphi \mathsf{U} \psi) & \Leftrightarrow & \exists k \in \mathsf{dom}(\lambda) \text{ s.t. } k > 0, \\
& & \big(\mathsf{M}, \lambda(k)\big) \models \psi, \text{and} \\
& & \forall h \in \mathsf{dom}(\lambda), \text{ if } 0 < h < k \\
& & \text{then } \big(\mathsf{M}, \lambda(h)\big) \models \varphi.
\end{array}$$

With the help of these modalities, we can elegantly represent the notions of potential boon and potential danger. In particular, a coalition C is said to be a potential boon for another coalition B, denoted $Boon(C, B)$, if C has a strategy that is beneficial for B. This can be defined as an abbreviation:

$$Boon(C, B) \stackrel{\mathsf{def}}{=} \langle\langle C{:}B{,}\emptyset\rangle\rangle\mathsf{X}\top.$$

Analogously, a coalition C is said to be a danger for another coalition H, denoted $Danger(C, H)$, if C has a strategy that is harmful for H. Formally:

$$Danger(C, H) \stackrel{\mathsf{def}}{=} \langle\langle C{:}\emptyset,H\rangle\rangle\mathsf{X}\top.$$

Notice we use the temporal operator G and U referring to the strict future, i.e., starting from the next state. But the temporal operators referring to the non-strict future can be defined as usual:

$$\langle\langle C{:}B,H\rangle\rangle\mathsf{G}^{\mathbf{ns}}\varphi \stackrel{\mathsf{def}}{=} \varphi \wedge \langle\langle C{:}B,H\rangle\rangle\mathsf{G}\varphi,$$
$$\langle\langle C{:}B,H\rangle\rangle(\varphi\mathsf{U}^{\mathbf{ns}}\psi) \stackrel{\mathsf{def}}{=} \psi \vee (\varphi \wedge \langle\langle C{:}B,H\rangle\rangle(\varphi\mathsf{U}\psi)).$$

also that standard ATL modalities $\langle\langle C\rangle\rangle\mathsf{X}\varphi$, $\langle\langle C\rangle\rangle\mathsf{G}^{\mathbf{ns}}\varphi$ and $\langle\langle C\rangle\rangle(\varphi\mathsf{U}^{\mathbf{ns}}\psi)$ are definable as special cases of $\mathsf{CA-ATL}$ modalities, as follows:

$$\langle\langle C\rangle\rangle\mathsf{X}\varphi \stackrel{\mathsf{def}}{=} \langle\langle C{:}\emptyset,\emptyset\rangle\rangle\mathsf{X}\varphi,$$
$$\langle\langle C\rangle\rangle\mathsf{G}^{\mathbf{ns}}\varphi \stackrel{\mathsf{def}}{=} \langle\langle C{:}\emptyset,\emptyset\rangle\rangle\mathsf{G}^{\mathbf{ns}}\varphi,$$
$$\langle\langle C\rangle\rangle(\varphi\mathsf{U}^{\mathbf{ns}}\psi) \stackrel{\mathsf{def}}{=} \langle\langle C{:}\emptyset,\emptyset\rangle\rangle(\varphi\mathsf{U}^{\mathbf{ns}}\psi).$$

For notational convenience, we denote with $\mathcal{L}_{\mathsf{ATL}}$ the ATL-fragment of the language $\mathcal{L}_{\mathsf{CA-ATL}}$ that only includes modalities of type $\langle\langle C{:}\emptyset,\emptyset\rangle\rangle$.

The second interpretation is based on what we call the weak satisfaction relation, denoted by $\models^{\mathtt{w}}$, which is defined analogously to the strong one, except that $StrSB^C_{\mathsf{M}}(s, B) \cap StrSH^C_{\mathsf{M}}(s, H)$ is replaced by $StrWB^C_{\mathsf{M}}(s, B) \cap StrWH^C_{\mathsf{M}}(s, H)$.

We conclude this section by defining two fragments of the language. The following syntaxs define the safety fragment ($\mathcal{L}_{\mathsf{CA-ATL^{-U}}}$) of $\mathcal{L}_{\mathsf{CA-ATL}}$:

$$\varphi, \psi ::= p \mid \neg\varphi \mid \varphi \wedge \psi \mid \langle\langle C{:}B,H\rangle\rangle\mathsf{X}\varphi \mid \langle\langle C{:}B,H\rangle\rangle\mathsf{G}\varphi$$

and the coalition logic fragment ($\mathcal{L}_{\mathsf{CA-CL}}$) of $\mathcal{L}_{\mathsf{CA-ATL}}$:

$$\varphi, \psi ::= p \mid \neg\varphi \mid \varphi \wedge \psi \mid \langle\langle C{:}B,H\rangle\rangle\mathsf{X}\varphi$$

where p ranges over $\mathbb{P}$ and C, B and H range over $2^{\mathbb{AGT}}$.

$\mathcal{L}_{\mathsf{CA-ATL}}$ and its fragments in combination with the satisfaction relation $\models$, and with the different CGMP classes defined in Section ??, give rise to a rich family of logics. In this paper, we focus on three logics from this family:

1. the one based on the safety fragment ($\mathcal{L}_{\mathsf{CA-ATL^{-U}}}$), and the model class $\mathbf{CP}^{\{\mathbf{WFP},\mathbf{CWFP}\}}$, which includes all CGMPs with well-founded and converse-well founded preferences;
2. the one based on the safety fragment ($\mathcal{L}_{\mathsf{CA-ATL^{-U}}}$), and the model class $\mathbf{CP}^{\{\mathbf{NFP},\mathbf{WFP},\mathbf{CWFP}\}}$, which includes all CGMPs with non-flat, well-founded and converse-well founded preferences;
3. the one based on the safety fragment ($\mathcal{L}_{\mathsf{CA-ATL^{-U}}}$), and the model class $\mathbf{CP}^{\{\mathbf{SP},\mathbf{WFP},\mathbf{CWFP}\}}$, which includes all CGMPs with stable, well-founded and converse-well founded preferences.

We consider two notions of validity for formulas in the language $\mathcal{L}_{\mathsf{CA-CL}}$. A formula $\varphi \in \mathcal{L}_{\mathsf{CA-CL}}$ is said to be valid, denoted by $\models \varphi$, if it is satisfied in every pointed CGMP under the $\models$ interpretation. Similarly, it is said to be **NFP**-valid, denoted by $\models^{\mathbf{NFP}} \varphi$, if it is satisfied in every pointed CGMP with non-flat preferences under the $\models$ interpretation.

6 Axiomatization

In this section, we introduce the logics Λ_1 and Λ_2, and prove their completeness with respect to their corresponding semantics.

6.1 Logic Λ_1

The following definition introduces the logic Λ_1.

Definition 11 (axiomatic system Λ_1). Λ_1 consists of the following axioms:

All tautologies of propositional logic (⊤)

$\neg\langle\langle C{:}B,H\rangle\rangle\mathsf{X}\bot$ (A-NAAA)

$\langle\langle C{:}B,H\rangle\rangle\mathsf{X}(\varphi \wedge \psi) \rightarrow \langle\langle C{:}B,H\rangle\rangle\mathsf{X}\varphi$ (A-MG)

$\langle\langle C{:}B,H\rangle\rangle\mathsf{X}\varphi \rightarrow \langle\langle C'{:}B',H'\rangle\rangle\mathsf{X}\varphi$ for $C \subseteq C', B \supseteq B', H \supseteq H'$ (A-MC)

$\langle\langle C : \emptyset, \emptyset\rangle\rangle\mathsf{X}\top$ (A-Ser)

$(\langle\langle C{:}B,H\rangle\rangle\mathsf{X}\varphi \wedge \langle\langle C'{:}B',H'\rangle\rangle\mathsf{X}\psi) \rightarrow$
$\langle\langle C \cup C'{:}B \cup B',H \cup H'\rangle\rangle\mathsf{X}(\varphi \wedge \psi)$ for $C \cap C' = \emptyset$ (A-Sup)

$\langle\langle C{:}B,H\rangle\rangle\mathsf{X}(\varphi \vee \psi) \rightarrow (\langle\langle C{:}B,H\rangle\rangle\mathsf{X}\varphi \vee \langle\langle \mathbb{AGT}{:}B,H\rangle\rangle\mathsf{X}\psi)$ (A-Cro)

$\langle\langle C\rangle\rangle\mathsf{G}\varphi \leftrightarrow \langle\langle C\rangle\rangle\mathsf{X}(\varphi \wedge \langle\langle C\rangle\rangle\mathsf{G}\varphi)$ (A-FP$_\mathsf{G}$)

$\langle\langle C{:}B,H\rangle\rangle\mathsf{G}\varphi \rightarrow \langle\langle C{:}B,H\rangle\rangle\mathsf{X}(\varphi \wedge \langle\langle C\rangle\rangle\mathsf{G}\varphi)$ (A-FP$_\mathsf{G}$)

$\langle\langle\emptyset\rangle\rangle\mathsf{G}\big(\theta \rightarrow \langle\langle C\rangle\rangle\mathsf{X}(\varphi \wedge \theta)\big) \rightarrow \langle\langle\emptyset\rangle\rangle\mathsf{G}(\theta \rightarrow \langle\langle C\rangle\rangle\mathsf{G}\varphi)$ (A-GFP$_\mathsf{G}$)

and the following rules of inference:

$$\frac{\varphi, \varphi \rightarrow \psi}{\psi} \quad \text{(MP)}$$

$$\frac{\varphi \leftrightarrow \psi}{\langle\langle C{:}B,H\rangle\rangle\mathsf{X}\varphi \leftrightarrow \langle\langle C{:}B,H\rangle\rangle\mathsf{X}\psi} \quad \text{(RE)}$$

$$\frac{\varphi}{\langle\langle C\rangle\rangle\mathsf{G}\varphi} \quad \text{(Gen)}$$

The names of axioms and inference rules reflect their intuitions. Axiom ?? is called no absurd available action, as its intuition is that a coalition's available joint action cannot ensure a logically absurd result. Axiom ?? is called monotonicity of goals: capabilities are monotonic with respect to goals. Similarly, Axiom ?? captures monotonicity of coalitions: capabilities are monotonic with respect to coalitions. Axiom ?? captures seriality. Axiom ?? captures superadditivity. Axiom ?? is the so-called crown axiom: it was called this way in (?) since it corresponds to the fact that the effectivity function of a game is a crown. The inference rules are modus ponens (??), replacement of equivalences (??) and generalization (??).

As usual, for every $\varphi \in \mathcal{L}_{\mathsf{CA-CL}}$, we write $\vdash \varphi$ to mean that φ is derivable in Λ_1, that is, there is a sequence of formulas $(\varphi_1, \ldots, \varphi_m)$ such that:

- $\varphi_m = \varphi$, and
- for every $1 \leq k \leq m$, either φ_k is an instance of one of the axiom schema of Λ_1 or there are formulas $\varphi_{k_1}, \ldots, \varphi_{k_t}$ such that $k_1, \ldots, k_t < k$ and $\frac{\varphi_{k_1}, \ldots, \varphi_{k_t}}{\varphi_k}$ is an instance of some inference rule of Λ_1.

In the axiomatic system Λ_1, the following formulas and inference rules are derivable.

Theorem 1. The following are derivable in Λ_1:

$$\frac{\varphi}{\langle\!\langle C \rangle\!\rangle \mathsf{G}^w \varphi} \qquad (\mathtt{Gen}_{\mathsf{G}^w})$$

$$\langle\!\langle \emptyset \rangle\!\rangle \mathsf{G}^w \big(\theta \to \langle\!\langle C \rangle\!\rangle \mathsf{X}(\varphi \wedge \theta)\big) \to \langle\!\langle \emptyset \rangle\!\rangle \mathsf{G}^w (\theta \to \langle\!\langle C \rangle\!\rangle \mathsf{G} \varphi) \qquad (\mathtt{th\text{-}GFP}_{\mathsf{G}})$$

$$\langle\!\langle \emptyset \rangle\!\rangle \mathsf{G}^w \big(\theta \to (\varphi \wedge \langle\!\langle C \rangle\!\rangle \mathsf{X}\theta)\big) \to \langle\!\langle \emptyset \rangle\!\rangle \mathsf{G}^w (\theta \to \langle\!\langle C \rangle\!\rangle \mathsf{G}^w \varphi) \qquad (\mathtt{th\text{-}GFP}_{\mathsf{G}^w})$$

$$\frac{\theta \to \langle\!\langle C \rangle\!\rangle \mathsf{X}(\varphi \wedge \theta)}{\theta \to \langle\!\langle C \rangle\!\rangle \mathsf{G}\varphi} \qquad (\mathtt{In}_{\mathsf{G}})$$

We are going to prove soundness and completeness of the logic Λ_1 relative to the notion of validity given at the end of Section ??. Our completeness proof uses the technique of canonical models under formula contexts. [6]

Theorem 2 (soundness and completeness). A formula $\varphi \in \mathcal{L}_{\mathsf{CA-CL}}$ is valid if and only if $\vdash \varphi$.

6.2 Logic Λ_2

The following definition introduces the logic Λ_2.

Definition 12 (Logic Λ_2). Λ_2 extends Λ_1 of Definition ?? with the following additional axiom schema:

$$\neg \langle\!\langle C{:}B,H \rangle\!\rangle \mathsf{X}\top \text{ for } B \cap H \neq \emptyset \qquad (\mathtt{A\text{-}BHC})$$

The additional axiom ?? is called the benefit–harm conflict. Its underlying intuition is that a coalition cannot simultaneously help and harm the same agent.

The definition of derivability is analogous to the one given in Section ?? for the logic Λ_1. We use $\vdash^{\mathbf{NFP}}$ to denote that φ is deducible in Λ_2. We now prove the soundness and completeness of the logic Λ_2 with respect to the notion of **NFP**-validity defined at the end of Section ??. The proof closely follows that of the logic Λ_1. Since Λ_2 is stronger than Λ_1, we can rely on the result of Lemma ??. The inductive step on the modal degree of formulas is also analogous. Therefore, it remains to prove the following downward validity and upward derivability lemma for Λ_2. Its proof is in Appendix B.2 of the supplementary material.

[6] add more details here or in the appendix?

Lemma 1 (Downward validity for Λ_2). Given a standard $\mathsf{CA-CL}$ disjunction

$$\varphi = \chi \vee \Big(\bigwedge_{x \in \mathbb{X}} \langle\!\langle C_x{:}B_x,H_x \rangle\!\rangle \mathsf{X}\psi_x \to \bigvee_{y \in \mathbb{Y}} \langle\!\langle C_y{:}B_y,H_y \rangle\!\rangle \mathsf{X}\psi_y\Big).$$

If $\models^{\mathbf{NFP}} \varphi$, then the following validity-reduction condition is satisfied:

- either $\models^{\mathbf{NFP}} \chi$;
- or there is $X \subseteq \mathbb{X}$ and $Y \subseteq \mathbb{Y}$ such that:
  - X is neat;
  - $\bigcup_{x\in X} C_x \subseteq C_y$, $\bigcup_{x\in X} B_x \supseteq B_y$, and $\bigcup_{x\in X} H_x \supseteq H_y$ for all $y \in \mathbb{Y}$;
  - there is at most one $y \in \mathbb{Y}$ such that $C_y \neq \mathbb{AGT}$;
  - and if $\bigcup_{x\in X} B_x \cap \bigcup_{x\in X} H_x = \emptyset$ then $\models^{\mathbf{NFP}} \bigwedge_{x\in X} \psi_x \to \bigvee_{y\in Y} \psi_y$.

The following is the upward derivability lemma for Λ_2. Its proof is in Appendix B.3 in the supplementary material.

Lemma 2 (Upward derivability for Λ_2). Given a standard $\mathsf{CA-CL}$ disjunction

$$\varphi = \chi \vee \Big(\bigwedge_{x \in \mathbb{X}} \langle\!\langle C_x{:}B_x,H_x \rangle\!\rangle \mathsf{X}\psi_x \to \bigvee_{y \in \mathbb{Y}} \langle\!\langle C_y{:}B_y,H_y \rangle\!\rangle \mathsf{X}\psi_y\Big).$$

We have $\vdash^{\mathbf{NFP}} \varphi$ if the following derivability-reduction condition is satisfied:

- either $\vdash \chi$,
- or there is $X \subseteq \mathbb{X}$ and $Y \subseteq \mathbb{Y}$ such that:
  - X is neat;
  - $\bigcup_{x\in X} C_x \subseteq C_y$, $\bigcup_{x\in X} B_x \supseteq B_y$, and $\bigcup_{x\in X} H_x \supseteq H_y$ for all $y \in \mathbb{Y}$;
  - there is at most one $y \in \mathbb{Y}$ such that $C_y \neq \mathbb{AGT}$;
  - and if $\bigcup_{x\in X} B_x \cap \bigcup_{x\in X} H_x = \emptyset$ then $\vdash^{\mathbf{NFP}} \bigwedge_{x\in X} \psi_x \to \bigvee_{y\in Y} \psi_y$.

7 Tree-like property and embedding

In the completeness proofs for the logics Λ_1 and Λ_2, we construct a tree model for every satisfiable formula. In this section, we show that satisfiability for the language $\mathcal{L}_{\mathsf{CA-ATL}}$ satisfies the tree-like model property. Based on this result, we provide polynomial embeddings of satisfiability for the language $\mathcal{L}_{\mathsf{CA-ATL}}$ into ATL-satisfiability, and of satisfiability for the language $\mathcal{L}_{\mathsf{CA-CL}}$ into CL-satisfiability. For the first embedding, we restrict to models with stable preferences. Thanks to these embeddings, we provide tight complexity results for satisfiability in our languages.

7.1 Tree-like model property

Let $\mathcal{R}^*$, $\mathcal{R}^-$ and $\mathcal{R}^+$ be, respectively, the reflexive and transitive closure, the inverse, and the transitive closure of $\mathcal{R}_\emptyset$.

Definition 13. Let $\mathsf{F} = (W, \mathbb{ACT}, \mathcal{R})$ be a CGF. We say that:

- M has a unique root iff there is a unique $w_0 \in W$ (called the root), such that, for every $v \in W$, $w_0 \mathcal{R}^* v$;
- M has unique predecessors iff for every $v \neq w_0$, the cardinality of $\mathcal{R}^-(v)$ is at most one;
- M has no cycles iff $\mathcal{R}^+$ is irreflexive;
- M is tree-like iff it has a unique root, unique predecessors and no cycles.

The property of "being tree-like" defined in Definition ??, is abbreviated as **TR**. This property naturally extends to CGMPs. We recall the notation introduced in Section ??: $\mathbf{CP}^X$, with $X \subseteq \{\mathbf{SP}, \mathbf{NFP}, \mathbf{TR}\}$, denotes the class of CGMPs satisfying all properties in X. The tree-like model property is stated in the following lemma, whose proof is given in Appendix C of the supplementary material.

Lemma 3. Let $\varphi \in \mathcal{L}_{\mathsf{CA-ATL}}$. Then,

- φ is satisfiable for the class $\mathbf{CP}^{\emptyset}$ iff φ is satisfiable for the class $\mathbf{CP}^{\{\mathbf{TR}\}}$,
- φ is satisfiable for the class $\mathbf{CP}^{\{\mathbf{SP}\}}$ iff φ is satisfiable for the class $\mathbf{CP}^{\{\mathbf{SP},\mathbf{TR}\}}$.

7.2 Embedding

Our embeddings rely on the following translation from $\mathcal{L}_{\mathsf{CA-ATL}}$ to the ATL fragment extended with special atomic formulas:

$$tr : \mathcal{L}_{\mathsf{CA-ATL}}(\mathbb{AGT}, \mathbb{P}) \longrightarrow \mathcal{L}_{\mathsf{ATL}}(\mathbb{AGT}, \mathbb{P}^+),$$

with $\mathbb{P}^+ = \mathbb{P} \cup \{\heartsuit_i : i \in \mathbb{AGT}\} \cup \{\triangle_i : i \in \mathbb{AGT}\}$:

$$\begin{aligned}
tr(p) &= p,\\
tr(\neg\varphi) &= \neg tr(\varphi),\\
tr(\varphi \wedge \psi) &= tr(\varphi) \wedge tr(\psi),\\
tr(\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\varphi) &= \langle\!\langle C\rangle\!\rangle \mathsf{X}\big(\heartsuit B \wedge \triangle H \wedge tr(\varphi)\big),\\
tr(\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}\varphi) &= \langle\!\langle C\rangle\!\rangle \mathsf{G}\big(\heartsuit B \wedge \triangle H \wedge tr(\varphi)\big),\\
tr\big(\langle\!\langle C{:}B,H\rangle\!\rangle (\varphi \mathsf{U} \psi)\big) &= \langle\!\langle C\rangle\!\rangle \big((\heartsuit B \wedge \triangle H \wedge tr(\varphi)) \mathsf{U} (\heartsuit B \wedge \triangle H \wedge tr(\psi))\big),
\end{aligned}$$

with $\heartsuit B \stackrel{\text{def}}{=} \bigwedge_{i \in B} \heartsuit_i$ and $\triangle H \stackrel{\text{def}}{=} \bigwedge_{i \in H} \triangle_i$. The special atomic formula $\heartsuit_i$ is read as "agent i is benefited," and $\triangle_i$ as "agent i is harmed". The idea behind the above translation is to label, at the language level, beneficial (resp. harmful) strategies of a coalition C for another coalition B (resp. H) with the special constructs $\heartsuit B$ (resp. $\triangle H$).

The key observation underlying the embedding for the full language $\mathcal{L}_{\mathsf{CA-ATL}}$ is that, in a tree-like model with stable preferences, whether a coalition's strategy is beneficial or harmful for an agent depends only on whether the next state is the second element of a best (resp. worst) computation for that agent. Stability of preferences is required to ensure that the $\heartsuit B/\triangle H$-labeling assigned at the current state remains consistent over time. For the embedding of the fragment $\mathcal{L}_{\mathsf{CA-CL}}$, stability of preferences is not required, since this language can refer only to the next state.

As the following theorem shows, by using the translation tr, satisfiability for the language $\mathcal{L}_{\mathsf{CA-CL}}$ with respect to the class $\mathbf{CP}^{\{\mathbf{SP}\}}$ is reducible to ATL-satisfiability, and satisfiability for the fragment $\mathcal{L}_{\mathsf{CA-CL}}$ with respect to $\mathbf{CP}^{\emptyset}$ is reducible to CL-satisfiability. Its proof is given in Appendix D of the supplementary material.

Theorem 3. Let $\varphi \in \mathcal{L}_{\mathsf{CA-ATL}}$. Then, φ is satisfiable for the class $\mathbf{CP}^{\{\mathbf{SP}\}}$ iff $tr(\varphi)$ is satisfiable for the class $\mathbf{CP}^{\{\mathbf{SP}\}}$.

Let $\varphi \in \mathcal{L}_{\mathsf{CA-CL}}$. Then, φ is satisfiable for the class $\mathbf{CP}^{\emptyset}$ iff $tr(\varphi)$ is satisfiable for the class $\mathbf{CP}^{\emptyset}$.

By Theorem ??, the fact that the size of $tr(\varphi)$ is polynomial, and the fact that satisfiability checking for ATL is Exptime-complete (?) and satisfiability checking for CL is Pspace-complete (?), we obtain the following tight complexity result.

Corollary 1. Checking satisfiability of formulas in $\mathcal{L}_{\mathsf{CA-ATL}}$ for the class $\mathbf{CP}^{\{SP\}}$ is Exptime-complete. It is Pspace-complete for both classes $\mathbf{CP}$ and $\mathbf{CP}^{\{\mathbf{SP}\}}$, when restricting to the fragment $\mathcal{L}_{\mathsf{CA-CL}}$.

8 Conclusion

We have introduced novel languages based on CL and ATL that enable reasoning about beneficial and harmful strategies, as well as the related notions of potential boon and potential danger. Our analysis is supported by several results concerning the axiomatizability and computational complexity of the proposed languages, obtained through embeddings into CL and ATL.

A number of questions remain unsolved, including the axiomatizability of the full ATL-based language $\mathcal{L}_{\mathsf{CA-ATL}}$ and the complexity of its satisfiability problem with respect to the class of CGMPs featuring possibly non-stable preferences. We plan to address these issues in future work. We also intend to extend our analysis to a risk-based notion of danger, according to which a coalition is a danger for another coalition if it has a strategy that, if chosen, with high probability brings harm to the latter. This will require extending CGMPs with a probabilistic component. Finally, we plan to combine the notion of danger investigated in this paper with a game-theoretic notion of rationality to define the concept of strong danger: a coalition C is a strong danger for another coalition H if it has a rational strategy that, if chosen, brings harm to H. It is called strong because it is realistic to assume that the agents in C may choose the harmful strategy, provided they are rational.

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A Proof of completeness for Λ_1

In this section, we show the proof of completeness for Λ_1 in details. The key step is constructing a model for any satisfiable standard conjunction, which equals to the negation of a standard disjunction. We use a method called 'blueprint' to achieve it.

A.1 Some derivabilities

Define $\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}^w\varphi = \varphi \wedge \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}\varphi$

Theorem 4. The following are derivable in Λ_1:

$$\frac{\varphi}{\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}^w\varphi} \quad (\mathtt{Gen}_{\mathsf{G}^w})$$

$$\langle\!\langle \emptyset\rangle\!\rangle \mathsf{G}^w\big(\theta \to \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}(\varphi\wedge\theta)\big) \to \langle\!\langle \emptyset\rangle\!\rangle \mathsf{G}^w(\theta \to \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}\varphi) \quad (\mathtt{th\text{-}GFP}_{\mathsf{G}})$$

$$\langle\!\langle \emptyset\rangle\!\rangle \mathsf{G}^w\big(\theta \to (\varphi\wedge \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\theta)\big) \to \langle\!\langle \emptyset\rangle\!\rangle \mathsf{G}^w(\theta \to \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}^w\varphi) \quad (\mathtt{th\text{-}GFP}_{\mathsf{G}^w})$$

$$\frac{\theta \to \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}(\varphi\wedge\theta)}{\theta \to \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}\varphi} \quad (\mathtt{In}_{\mathsf{G}})$$

B Canonical method

Definition 14 (formula context). Given an formula ψ, denote the set of all subformulas of ψ by $sub(\psi)$. Let $sub^+(\psi) = \{\top,\bot\} \cup sub(\psi) \cup \{\neg\varphi \mid \varphi \in sub(\psi)\}$, $cl(\psi) = sub^+(\psi) \cup \{\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\varphi \mid C \subseteq \mathbb{AGT}, B \subseteq \mathbb{AGT}, H \subseteq \mathbb{AGT}, \varphi \in sub^+(\psi)\}$, $cl^+(\psi) = cl(\psi) \cup \{\neg\varphi \mid \varphi \in cl+(\psi)\}$, and $ecl(\psi) = cl^+(\psi) \cup \{\bigvee_{s\in\Gamma}\bigwedge s \mid \Gamma \subseteq 2^{cl^+(\psi)}\}$

Definition 15 (Maximal consistent sets). We say a set $s \subseteq ecl(\psi)$ of formnulas is maximal consistent sets under ψ-context iff s is consistent and for all $s^+ \subseteq ecl(\psi)$ if $s \subset s^+$ then s^+ is inconsistent.

Maximal consistent sets under ψ-context is called canonical states under ψ-context. The set of all maximal consistent sets under ψ-context by $\mathbb{S}^\psi$.

Definition 16 (Characteristic formula). Let $\Omega \subseteq \mathbb{S}^\psi$ and $\chi \in ecl(\varphi)$, we say χ is a characteristic formula of Ω iff for all $s \subseteq \mathbb{S}^\psi$ we have $s \vdash \chi$ iff $s \in \Omega$.

Theorem 5. Every $\Omega \subseteq \mathbb{S}^\psi$ has a characteristic formula.

Definition 17 (canonical deterministic action). Given an formula ψ, define

$$\mathtt{LP} = \{\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\chi \mid \chi \in ecl(\psi)\}$$

$$\mathtt{CF} = \{\mathtt{cf} : 2^{\mathbb{S}^\psi}/\{\emptyset\} \to \mathbb{S}^\psi \mid (\forall \Omega \in 2^{\mathbb{S}^\psi}/\{\emptyset\})\mathtt{cf}(\Omega) \in \Omega\}$$

$$\mathbb{ACT}^\psi = \mathtt{LP} \times \mathbb{AGT} \times \mathtt{CF}$$

where $\mathtt{LP}$ is the set of local powers, $\mathtt{CF}$ the set of choice functions, and $\mathbb{ACT}^\psi$ is the set of canonical deterministic actions.

Definition 18 (support and consesus). For every canonical deterministic action $(\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\chi, a, \mathtt{cf}) \in \mathbb{ACT}^\psi$, define $\mathtt{support}((\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\chi, a, \mathtt{cf})) = \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\chi$. For every aciton profile $\delta : \mathbb{AGT} \to \mathbb{ACT}^\psi$, define $\mathtt{CON}(\delta) = \{\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\chi \in \mathtt{LP} \mid (\forall a \in C)\mathtt{support}(\delta(a)) = \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\chi\}$. The set $\mathtt{CON}(\delta)$ is called consesus of agents on action profile δ.

Lemma 4. For all aciton profile δ, we have:

- $\langle\!\langle \emptyset\rangle\!\rangle \mathsf{X}\top \in \mathtt{CON}(\delta)$.
- For all $\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\chi, \langle\!\langle C'{:}B',H'\rangle\!\rangle \mathsf{X}\chi' \in \mathtt{CON}(\delta)$ with $\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\chi \neq \langle\!\langle C'{:}B',H'\rangle\!\rangle \mathsf{X}\chi'$, then $C \cap C' = \emptyset$.

Proof. The first item is obvious. For the second item, suppose that there are $\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\chi, \langle\!\langle C'{:}B',H'\rangle\!\rangle \mathsf{X}\chi' \in \mathtt{CON}(\delta)$ with $\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\chi \neq \langle\!\langle C'{:}B',H'\rangle\!\rangle \mathsf{X}\chi'$ and $C \cap C' \neq \emptyset$. Then there is an agent $a \in C \cap C'$ such that $\mathtt{support}(\delta(a)) = \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\chi$ and $\mathtt{support}(\delta(a)) = \langle\!\langle C'{:}B',H'\rangle\!\rangle \mathsf{X}\chi'$, which contradicts to the fact that $\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\chi \neq \langle\!\langle C'{:}B',H'\rangle\!\rangle \mathsf{X}\chi'$. □

Definition 19 (Candidate successor). For every canonical states $s \in \mathbb{S}^\psi$ and every aciton profile $\delta : \mathbb{AGT} \to \mathbb{ACT}^\psi$, define s' is a candidate successors of δ at s, iff:

- for all $\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\chi \in \mathtt{CON}(\delta)$, if $s \vdash \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\chi$, then $s' \vdash \chi$;
- for all $\chi \in ecl(\psi)$, if $s \nvdash \langle\!\langle \mathbb{AGT}{:}\emptyset,\emptyset\rangle\!\rangle \mathsf{X}\chi$, then $s' \vdash \neg\chi$.

The set of all candidate successors of δ at s is denoted by $\mathtt{CS}(s,\delta)$.

The above definition hint that the transition relation in the canonical model is defined based on the candidate successors. Note $\langle\!\langle \emptyset\rangle\!\rangle \mathsf{X}\top \in s$ and $\langle\!\langle \emptyset\rangle\!\rangle \mathsf{X}\top\mathtt{CON}(\delta)$. Then $\mathtt{CS}(s,\delta)$ is non-empty andthe transition relation can be defined with seriality. However, the set of candidate successors may be not singleton. We need some mechanism to select a unique successor from the candidate successors. The tricky election defined below serves this purpose.

Definition 20 (voting and tricky election). For every canonical deterministic action $(\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\chi, a, \mathtt{cf}) \in \mathbb{ACT}^\psi$, define $\mathtt{vote}((\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\chi, a, \mathtt{cf})) = a$.

For every aciton profile $\delta : \mathbb{AGT} \to \mathbb{ACT}^\psi$, define $\mathtt{te}(\delta) = (\sum_{a\in\mathbb{AGT}} \mathtt{vote}(\delta(a))) \bmod n$, where $n = \|\mathbb{AGT}\|$. The agent $\mathtt{te}(\delta)$ is called the electee of tricky election on action profile δ.

Definition 21 (decision). For every canonical deterministic action $(\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\chi, a, \mathtt{cf}) \in \mathbb{ACT}^\psi$, define

$$\mathtt{de}((\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\chi, a, \mathtt{cf})) = \mathtt{cf}$$

Definition 22 (canonical model without preferences). Given a formula ψ, define the following structure $\mathsf{M}^\psi = (\mathbb{S}^\psi, \mathbb{ACT}^\psi, \mathcal{R}^\psi, \mathcal{V}^\psi)$:

- $\mathbb{S}^\psi$ is the set of all canonical states under ψ-context;
- $\mathbb{ACT}^\psi$ is the set of all canonical deterministic actions under ψ-context;
- $\mathcal{R}^\psi$ assigns to every action profile a a transition relation on $\mathbb{S}$ such that for all $s, s' \in \mathbb{S}^\psi$ and $\delta : \mathbb{AGT} \to \mathbb{ACT}^\psi$, we have $s\mathcal{R}_\delta s'$ iff $s' = \mathtt{de}(\delta(a))(\mathtt{CS}(s,\delta))$, where $a = \mathtt{te}(\delta)$;
- $\mathcal{V}^\psi(p) = \{s \in \mathbb{S}^\psi \mid p \in s\}$ for all propositional variable p in $ecl(\varphi)$.

Lemma 5 (Existence lemma). For all canonical state $s \in \mathbb{S}^\psi$, $\langle\!\langle C_1\rangle\!\rangle \mathsf{X}\chi_1, \ldots, \langle\!\langle C_k\rangle\!\rangle \mathsf{X}\chi_k \in ecl(\psi)$ such that $s \vdash \langle\!\langle C_x\rangle\!\rangle \mathsf{X}\chi_x$ for all $1 \leq x \leq k$, and $\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\chi \in ecl(\psi)$ with $s \nvdash \langle\!\langle C_x\rangle\!\rangle \mathsf{X}\chi_x$, if $C_i \cap C_j = \emptyset$ for all $1 \leq i \neq j \leq k$ and $C_i \subseteq C$ for all $1 \leq i \leq k$, there is a action profile δ and a state s' such that $s\mathcal{R}_\delta s'$ and $s' \vdash \chi_i$ for all $1 \leq i \leq k$ and $s' \vdash \neg\chi$.

Theorem 6 (Local Realization lemma). For every coaltition C, canonical state $s \in \mathbb{S}^\psi$, and $\chi \in ecl(\psi)$, we have: $s \vdash \langle\!\langle C\rangle\!\rangle \mathsf{X}\chi$ iff there is a joint action δ_C of C such that $\mathcal{F}_C$ is available at s and $\mathcal{R}_{\delta_C}(s) \subseteq |\chi|$.

Proof. Fix coaltition C, canonical state $s \in \mathbb{S}^\psi$, and $\Omega \subseteq \mathbb{S}^\psi$. Let χ be a characteristic formula of Ω. From left to right, assume $s \vdash \langle\!\langle C\rangle\!\rangle \mathsf{X}\chi$. Let $\delta_C : C \to \mathbb{ACT}^\psi$ such that $\mathtt{support}(\delta_C(a)) = \langle\!\langle C\rangle\!\rangle \mathsf{X}\chi$. Then $\langle\!\langle C\rangle\!\rangle \mathsf{X}\chi \in \mathtt{CON}(\delta_{\mathbb{AGT}})$ for all $\delta_{\mathbb{AGT}} : \mathbb{AGT} \to \mathbb{ACT}^\psi$ with $\delta_C \subseteq \delta_{\mathbb{AGT}}$. Note there is $\delta_{\mathbb{AGT}} : \mathbb{AGT} \to \mathbb{ACT}^\psi$ such that $\delta_C \subseteq \delta_{\mathbb{AGT}}$, $\mathtt{support}(\delta_{\mathbb{AGT}}(b)) = \langle\!\langle \emptyset\rangle\!\rangle \mathsf{X}\top$ for all $b \in \mathbb{AGT} - C$ and $\delta_{\mathbb{AGT}}$ is available at s. Then δ_C is available at s. Let $\delta_{\mathbb{AGT}} : \mathbb{AGT} \to \mathbb{ACT}^\psi$ with $\delta_C \subseteq \delta_{\mathbb{AGT}}$. Note $\langle\!\langle C\rangle\!\rangle \mathsf{X}\chi \in \mathtt{CON}(\delta_{\mathbb{AGT}})$ and $s \vdash \langle\!\langle C\rangle\!\rangle \mathsf{X}\chi$. Then $s' \vdash \chi$ for all $s' \in \mathtt{CS}(s, \delta_{\mathbb{AGT}})$. Then $s' \in \Omega$ for all $s' \in \mathcal{R}_{\delta_C}(s)$.

From right to left, assume $s \nvdash \langle\!\langle C\rangle\!\rangle \mathsf{X}\chi$. Let δ_C be a joint action of C such that δ_C is available at s. Let $\delta_{\mathbb{AGT}} : \mathbb{AGT} \to \mathbb{ACT}^\psi$ such that $\delta_C \subseteq \delta_{\mathbb{AGT}}$, $\mathtt{support}(\delta_{\mathbb{AGT}}(b)) = \langle\!\langle \emptyset\rangle\!\rangle \mathsf{X}\top$ for all $b \in \mathbb{AGT} - C$. Then $\mathtt{CON}(\delta_{\mathbb{AGT}}) = \mathtt{CON}(\delta_C)$. Note $s \nvdash \langle\!\langle C\rangle\!\rangle \mathsf{X}\chi$. Then $\{\langle\!\langle A\rangle\!\rangle \mathsf{X}\varphi \in \mathtt{CON}(\delta_C) \mid s \vdash \langle\!\langle A\rangle\!\rangle \mathsf{X}\varphi\} \cup \{\neg\langle\!\langle C\rangle\!\rangle \mathsf{X}\chi\}$ is consistent. We want to show $\{\varphi \mid \langle\!\langle A\rangle\!\rangle \mathsf{X}\varphi \in \mathtt{CON}(\delta_C), s \vdash \langle\!\langle A\rangle\!\rangle \mathsf{X}\varphi\} \cup \{\neg\chi\}$ is consistent. It suffices to show that if $\vdash \bigwedge_{\langle\!\langle A\rangle\!\rangle \mathsf{X}\varphi \in \mathtt{CON}(\delta_C), s \vdash \langle\!\langle C\rangle\!\rangle \mathsf{X}\varphi} \varphi \to \chi$ then $\vdash \bigwedge_{\langle\!\langle A\rangle\!\rangle \mathsf{X}\varphi \in \mathtt{CON}(\delta_C), s \vdash \langle\!\langle C\rangle\!\rangle \mathsf{X}\varphi} \langle\!\langle A\rangle\!\rangle \mathsf{X}\varphi \to \langle\!\langle C\rangle\!\rangle \mathsf{X}\chi$. It's clear by supperadditivity. Therefore, there is $s' \in \mathbb{S}^\psi$ such that $\{\varphi \mid \langle\!\langle A\rangle\!\rangle \mathsf{X}\varphi \in \mathtt{CON}(\delta_C), s \vdash \langle\!\langle A\rangle\!\rangle \mathsf{X}\varphi\} \cup \{\neg\chi\} \subseteq s'$ and $s\mathcal{R}_{\delta_{\mathbb{AGT}}} s'$. Note $\chi \notin s'$. Then $s' \in \Omega$. ☐

Theorem 7 (G-Realization Lemma). For $s \in \mathbb{S}^\psi$ and $\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}\varphi \in ecl(\varphi)$, we have $\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}\varphi \in s$ iff there is a C-strategy $\mathcal{F}_C$ such that for all $\lambda \in \mathcal{O}(s, \mathcal{F}_C)$ and all $i > 0$, we have $\varphi \in \lambda[i]$.

Proof. Fix $s \in \mathbb{S}^\psi$ and $\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}\varphi \in ecl(\varphi)$. From left to right, suppose that $\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}\varphi \in s$. By axiom ?? and Local Realization lemma, for all $s' \in |\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}\varphi|$, there is a joint action δ_C such that $\mathcal{R}_{\delta_C}(s') \subseteq |\varphi \wedge \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}\varphi|$. Let $\mathcal{F}_C$ be a C-strategy $\mathcal{F}_C$ such that for all $s' \in |\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}\varphi|$ we have $\mathcal{F}_C(s)$ is a joint action δ_C such that $\mathcal{R}_{\delta_C}(s') \subseteq |\varphi \wedge \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}\varphi|$. Let $\lambda \in \mathcal{O}(s, \mathcal{F}_C)$, by induction on $i > 0$, we can show that $\varphi \in \lambda[i]$.

From right to left, suppose that there is a C-strategy $\mathcal{F}_C$ such that $\varphi \in \lambda[i]$ for all $\lambda \in \mathcal{O}(s, \mathcal{F}_C)$ and all $i > 0$. Let $\Omega = \{s\} \cup \{\lambda[i] \mid \lambda \in \mathcal{O}(s, \mathcal{F}_C), i > 0\}$ and χ be the characteristic formula of Ω. Note $\mathcal{R}_{\mathcal{F}_C(s')}(s') \subseteq \Omega$ for all $s' \in \Omega$. Then $s' \vdash \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}(\varphi \wedge \chi)$ for all $s' \in \Omega$. Note $\chi \in s$. It suffices to show that $\vdash \chi \to \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}\varphi$. By ??, it suffices to show that $\vdash \chi \to \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}(\varphi \wedge \chi)$. Let s^+ be a maximal consistent set such that $\chi \in s^+$. Note $s^+ \cap ecl(\varphi) \in \mathbb{S}^\psi$. Then $s^+ \cap ecl(\varphi) \in \Omega$. Therefore, $s^+ \cap ecl(\varphi) \vdash \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}(\varphi \wedge \chi)$. Therefore, $\vdash \chi \to \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}(\varphi \wedge \chi)$. Therefore, $\vdash \chi \to \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}\varphi$ and $\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}\varphi \in s$. ☐

C Proof of completeness for Λ_2

In this section, we show the proof of of completeness for Λ_2 in details. The approach is similar to section ??. However, we need to construct models with non-flat preferences now. For convenience, we will simply write **NFP**-satisfiable for "satisfiable on a non flat model".

C.1 benefit-harm conflict blueprints and their non-flat realizations

To build the models with non-flat preferences, we need the blueprints satisfying the property "benefit-harm conflict".

Definition 23 (Benefit-harm conflict blueprint). A blueprint $\mathtt{bp} = (\mathbb{ACT}_0, \mathtt{clist}, \mathtt{Blist}, \mathtt{Hlist})$ is benefit-harm conflict, iff for all $\delta \in \mathbb{ACT}_0^{\mathbb{AGT}}$:

- $\mathtt{Blist}(\delta) \cap \mathtt{Hlist}(\delta) = \emptyset$;
- and $\mathtt{clist}(\delta)$ is **NFP**-satisfiable.

Theorem 8 (Non-flat realization). Given a benefit-harm conflict blueprint $\mathtt{bp} = (\mathbb{ACT}_0, \mathtt{clist}, \mathtt{Blist}, \mathtt{Hlist})$ and a **NFP**-satisfiable conjunction of propositional literals χ. Then there is a pointed non-flat model (M, s_0), called a non-flat realization of $\mathtt{bp}$ and χ, where $\mathsf{M} = (\mathsf{F}, \preceq, \mathcal{V})$, $\mathsf{F} = (\mathbb{S}, \mathbb{ACT}, \mathcal{R})$, and $\mathcal{R} = (\mathcal{R}_\delta)_{\delta \in \mathbb{JACT}}$ such that:

- $\mathsf{M}, s_0 \models \chi$;
- $(\mathbb{ACT}_0 \times \mathbb{Z}/\{0\}) \subseteq \mathbb{ACT}$;
- for all $\mathcal{F}_{\mathbb{AGT}} \in Str_M^{\mathbb{AGT}}$ and $\delta : \mathbb{AGT} \to \mathbb{ACT}_0$, if for all $a \in \mathbb{AGT}$ there is $z \in \mathbb{Z}$ such that $\mathcal{F}(s_0)(a) = (\delta(a), z)$, then $(\mathsf{M}, \wr_{\mathsf{M}}(s_0, \mathcal{F}_{\mathbb{AGT}})(1)) \models \mathtt{clist}(\delta)$;
- for all $\mathcal{F}_{\mathbb{AGT}} \in Str_M^{\mathbb{AGT}}$ and $\delta : \mathbb{AGT} \to \mathbb{ACT}_0$, if for all $a \in \mathbb{AGT}$ there is $z \in \mathbb{Z}$ such that $\mathcal{F}(s_0)(a) = (\delta(a), z)$, then for all $i \in \mathbb{AGT}$, $\wr_{\mathsf{M}}(s_0, \mathcal{F}_{\mathbb{AGT}}) \in Best_{i,s_0}$ iff $i \in \mathtt{Blist}(\delta)$, and $\wr_{\mathsf{M}}(s_0, \mathcal{F}_{\mathbb{AGT}}) \in Worst_{i,s_0}$ iff $i \in \mathtt{Hlist}(\delta)$.

Proof. For all $\delta \in \mathbb{ACT}_0^{\mathbb{AGT}}$, let $(\mathsf{M}^\delta, s^\delta)$ be a pointed model satisfies $\mathtt{clist}(\delta)$, where $\mathsf{M} = (\mathbb{S}^\delta, \mathbb{ACT}^\delta, \mathcal{R}^\delta, \preceq^\delta, \mathcal{V}^\delta)$, their domains are pairwise disjoint, and their set of actions are pairwise disjoint and disjoint with $\mathbb{ACT}_0$. Let s_0 be three states not in any of these models.

Before constructing the model, we first define the pseudo-utility function $\mathcal{PU} : \{\delta : \mathbb{AGT} \to (\mathbb{ACT}_0 \times \mathbb{Z}/\{0\})\} \to \mathbb{Z}/\{0\}$ satisfying for all $\delta : \mathbb{AGT} \to (\mathbb{ACT}_0 \times \mathbb{Z}/\{0\})$:

$$\mathcal{PU}(\delta) = \sum_{a \in \mathbb{AGT}} second(\delta(a))$$

where $second(\delta(a))$ is the second element of the pair $\delta(a)$ for every $a \in \mathbb{AGT}$.

Define the CGMP $\mathsf{M} = (\mathbb{S}, \mathbb{ACT}, \mathcal{R}, \preceq, \mathcal{V})$ as the following:

- $\mathbb{S} = \{s_0\} \cup \bigcup_{\delta \in \mathbb{ACT}_0^{\mathbb{AGT}}} (\mathbb{S}^\delta \times \mathbb{Z}/\{0\})$;
- $\mathbb{ACT} = (\mathbb{ACT}_0 \times \mathbb{Z}/\{0\}) \cup \bigcup_{\delta \in \mathbb{ACT}_0^{\mathbb{AGT}}} \mathbb{ACT}^\delta$;
- for all $s, s' \in \mathbb{S}$ and $\delta : \mathbb{AGT} \to \mathbb{ACT}$: $s\mathcal{R}_\delta s'$ iff one of the following condition is satisfied:
  - there is $\sigma \in \mathbb{ACT}_0^{\mathbb{AGT}}$, $z \in \mathbb{Z}/\{0\}$ and $u, v \in St^\sigma$ such that $s = (u, z)$, $s' = (v, z)$, $\delta : \mathbb{AGT} \to \mathbb{ACT}^\sigma$ and $u\mathcal{R}^\sigma_\delta v'$;
  - $s = s_0$ $\delta : \mathbb{AGT} \to (\mathbb{ACT}_0 \times \mathbb{Z}/\{0\})$, there is $\sigma : \mathbb{AGT} \to \mathbb{ACT}_0$ such that $first(\delta(a)) = \sigma(a)$ for all $a \in \mathbb{AGT}$ and $s' = (s^\sigma, \mathcal{PU}(\delta))$, where $first(\delta(a))$ is the first element of the pair $\delta(a)$ for every $a \in \mathbb{AGT}$;
- for all $s \in \mathbb{S}$:

$$\mathcal{V}(s) = \begin{cases} \mathcal{V}^\delta(u) & \text{if there is } \delta \in \mathbb{ACT}_0^{\mathbb{AGT}} \\ & \text{and } z \in \mathbb{Z}/\{0\} \text{ such that } s = (u, z), \\ \{p \in \mathbb{P} \mid p \text{ is a conjunct of } \chi\} & \text{if } s = s_0 \end{cases}$$

- for all $a \in \mathbb{AGT}$ and $s \in \mathbb{S}$:

$$\preceq_{a,s} = \begin{cases} \preceq^\delta_{a,s} & \text{if } s \in \mathbb{S}^\delta \text{ for some } \delta \in \mathbb{ACT}_0^{\mathbb{AGT}} \\ \preceq^0_a & \text{if } s = s_0 \end{cases}$$

  where $\preceq^0_a = \{(\lambda, \lambda') \in Comp_{\mathsf{M},s_0} \times Comp_{\mathsf{M},s_0} \mid U_a(\lambda) \leq U_a(\lambda')\}$, $U : Comp_{\mathsf{M},s_0} \to \mathbb{Z}$ is the utility function of a, such that for all $\lambda \in Comp_{\mathsf{M},s_0}$:
  - if there is $\delta : \mathbb{AGT} \to (\mathbb{ACT}_0 \times \mathbb{Z})$ and $\sigma \in \mathbb{ACT}_0^{\mathbb{AGT}}$ such that $first(\delta(a)) = \sigma(a)$, $a \notin \mathtt{Blist}(\sigma)$, $s_0\mathcal{R}_\delta\lambda(1)$ and $\mathcal{PU}(\delta) > 0$, then $U_a(\lambda) = \mathcal{PU}(\delta)$;
  - if there is $\delta : \mathbb{AGT} \to (\mathbb{ACT}_0 \times \mathbb{Z})$ and $\sigma \in \mathbb{ACT}_0^{\mathbb{AGT}}$ such that $first(\delta(a)) = \sigma(a)$, $a \notin \mathtt{Hlist}(\sigma)$, $s_0\mathcal{R}_\delta\lambda(1)$ and $\mathcal{PU}(\delta) < 0$, then $U_a(\lambda) = \mathcal{PU}(\delta)$;
  - if not the above case and $U_a(\lambda) > 0$, then $U_a(\lambda) = 1$;
  - otherwise $U_a(\lambda) = -1$.

It is clear that $\mathsf{M}, s_0 \models \chi$ and $\mathbb{ACT}_0 \subseteq \mathbb{ACT}$.

Let $\mathcal{F} \in Str_M^{\mathbb{AGT}}$ and $\delta : \mathbb{AGT} \to \mathbb{ACT}_0$ such that $\mathcal{F}(a)(s_0) = \delta(a)$ for all $a \in \mathbb{AGT}$. We want to show $M, \wr_M(s_0, \mathcal{F})(1) \models \mathtt{clist}(\delta)$. Note $\mathcal{R}_\delta(s_0) = \{s^\delta\}$. Then $\wr_M(s_0, \mathcal{F})(1) = s^\delta$. Therefore,

$$M, \wr_M(s_0, \mathcal{F})(1) \models \mathtt{clist}(\delta).$$

Let $\mathcal{F}_{\mathbb{AGT}} \in Str_M^{\mathbb{AGT}}$ and $\delta : \mathbb{AGT} \to \mathbb{ACT}_0$ such that $a \in \mathbb{AGT}$ there is $z \in \mathbb{Z}$ such that $\mathcal{F}(s_0)(a) = (\delta(a), z)$. We want to show for all $i \in \mathbb{AGT}$, $\wr_{\mathsf{M}}(s_0, \mathcal{F}_{\mathbb{AGT}}) \in Best_{i,s_0}$ iff $i \in \mathtt{Blist}(\delta)$, and $\wr_{\mathsf{M}}(s_0, \mathcal{F}_{\mathbb{AGT}}) \in Worst_{i,s_0}$ iff $i \in \mathtt{Hlist}(\delta)$. Let $i \in \mathbb{AGT}$. We only show that $\wr_{\mathsf{M}}(s_0, \mathcal{F}_{\mathbb{AGT}}) \in Best_{i,s_0}$ iff $i \in \mathtt{Blist}(\delta)$ here. The proof for $\wr_{\mathsf{M}}(s_0, \mathcal{F}_{\mathbb{AGT}}) \in Worst_{i,s_0}$ iff $i \in \mathtt{Hlist}(\delta)$ is similar.

From left to right, assume $\wr_{\mathsf{M}}(s_0, \mathcal{F}_{\mathbb{AGT}}) \in Best_{i,s_0}$. Then there is no $\lambda \in Comp_{\mathsf{M},s_0}$ such that $U_i(\lambda) > U_i(\wr_{\mathsf{M}}(s_0, \mathcal{F}_{\mathbb{AGT}}))$. Note for all $k \in \mathbb{N}$ there is always some $\sigma : \mathbb{AGT} \to (\mathbb{ACT}_0 \times \mathbb{Z})$ such that $first(\sigma(i)) = \delta(i)$,, $s_0\mathcal{R}_\delta\lambda(1)$ and $\mathcal{PU}(\sigma) > 0$. If $i \notin \mathtt{Blist}$ then there is always some $\lambda \in Comp_{\mathsf{M},s_0}$ such that $U_i(\lambda) > U_i(\wr_{\mathsf{M}}(s_0, \mathcal{F}_{\mathbb{AGT}}))$. Therefore, $i \in \mathtt{Blist}(\delta)$.

From right to left, assume $i \in \mathtt{Blist}(\delta)$. Then $U_i(\lambda) \leq 1$ for all $\lambda \in Comp_{\mathsf{M},s_0}$. Note $U_i(\lambda) = 1$ for all $\lambda \in \mathcal{O}_{\mathsf{M}}(s_0, \mathcal{F}_{\mathbb{AGT}})$. Therefore, $\wr_{\mathsf{M}}(s_0, \mathcal{F}_{\mathbb{AGT}}) \in Best_{i,s_0}$. □

C.2 The proof of downward validity for Λ_2

Now we prove the downward validity for Λ_2(Lemma ??).

Proof. We prove its contrapositive. Assume validity-reduction-condition is not met. Then it meets satisfiability-reduction-condition, that is: $\neg\chi$ is satisfiable and for all $X \subseteq \mathbb{X}$ and $Y \subseteq \mathbb{Y}$, if

- X is neat;
- $\bigcup_{x \in X} C_x \subseteq C_y$, $\bigcup_{x \in X} B_x \supseteq B_y$, and $\bigcup_{x \in X} H_x \supseteq H_y$ for all $y \in \mathbb{Y}$;
- and there is at most one $y \in \mathbb{Y}$ such that $C_y \neq \mathbb{AGT}$;

then $\bigcup_{x \in X} B_x \cap \bigcup_{x \in X} H_x = \emptyset$ and $\bigwedge_{x \in X} \psi_x \wedge \bigwedge_{y \in Y} \neg\psi_y$ is **NFP**-satisfiable.

We want to show that the following formula is satisfiable:

$$\neg\chi \wedge \bigwedge_{x \in \mathbb{X}} \langle\!\langle C_x{:}B_x{,}H_x \rangle\!\rangle \mathsf{X}\psi_x \wedge \bigwedge_{y \in \mathbb{Y}} \neg\langle\!\langle C_y{:}B_y{,}H_y \rangle\!\rangle \mathsf{X}\psi_y$$

Note φ is satisfiable iff $\varphi \wedge \neg\langle\!\langle \mathbb{AGT}{:}\emptyset{,}\emptyset \rangle\!\rangle \mathsf{X}\bot$ is satisfiable for all formula φ. Therefore, for convenience, we can assume $\mathbb{Y}$ is a nonempty beginning segment of natural numbers.

The construction of $\mathtt{bp} = (\mathbb{ACT}_0, \mathtt{clist}, \mathtt{Blist}, \mathtt{Hlist})$ is the same with ??. We want to show $\mathtt{bp}$ is benefit-harm conflict. Similar to the proof in section ??, $\mathtt{clist}(\delta)$ is **NFP**-satisfiable for all $\delta \in \mathbb{ACT}_0^{\mathbb{AGT}}$. Note $\mathtt{Blist}(\delta) = \bigcup_{x \in \mathtt{support}(\delta)} B$, $\mathtt{Hlist}(\delta) = \bigcup_{x \in \mathtt{support}(\delta)} H$, and $\bigcup_{x \in \mathtt{support}(\delta)} B_x \cap \bigcup_{x \in \mathtt{support}(\delta)} H_x = \emptyset$. Then $\mathtt{Blist}(\delta) \cap \mathtt{Hlist}(\delta) = \emptyset$ for all $\delta \in \mathbb{ACT}_0^{\mathbb{AGT}}$. Therefore, $\mathtt{bp}$ is a benefit-harm conflict blueprint.

Let $\overline{\chi}$ be a conjunction of atomic propositions, equivalent to $\neg\chi$. By non-flat realization theorem (theorem ??), there is a pointed non-flat model (M, s_0), such that:

- $\mathsf{M}, s_0 \models \overline{\chi}$;
- $(\mathbb{ACT}_0 \times \mathbb{Z}) \subseteq \mathbb{ACT}$;
- for all $\mathcal{F}_{\mathbb{AGT}} \in Str_M^{\mathbb{AGT}}$ and $\delta : \mathbb{AGT} \to \mathbb{ACT}_0$, if for all $a \in \mathbb{AGT}$ there is $z \in \mathbb{Z}$ such that $\mathcal{F}(s_0)(a) = (\delta(a), z)$, then $(\mathsf{M}, \wr_{\mathsf{M}}(s_0, \mathcal{F}_{\mathbb{AGT}})(1)) \models \mathtt{clist}(\delta)$;
- for all $\mathcal{F}_{\mathbb{AGT}} \in Str_M^{\mathbb{AGT}}$ and $\delta : \mathbb{AGT} \to \mathbb{ACT}_0$, if for all $a \in \mathbb{AGT}$ there is $z \in \mathbb{Z}$ such that $\mathcal{F}(s_0)(a) = (\delta(a), z)$, then for all $i \in \mathbb{AGT}$, $\wr_{\mathsf{M}}(s_0, \mathcal{F}_{\mathbb{AGT}}) \in Best_{i,s_0}$ iff $i \in \mathtt{Blist}(\delta)$, and $\wr_{\mathsf{M}}(s_0, \mathcal{F}_{\mathbb{AGT}}) \in Worst_{i,s_0}$ iff $i \in \mathtt{Hlist}(\delta)$.

Similar to the proof in section ??, it can be verified that

$$\mathsf{M}, s_0 \models \neg\chi \wedge \bigwedge_{x \in \mathbb{X}} \langle\!\langle C_x{:}B_x{,}H_x \rangle\!\rangle \mathsf{X}\psi_x \wedge \bigwedge_{y \in \mathbb{Y}} \neg\langle\!\langle C_y{:}B_y{,}H_y \rangle\!\rangle \mathsf{X}\psi_y.$$

□

C.3 The proof of upward derivability for Λ_2

Now we can prove the upward derivability lemma for Λ_2.

Proof. Assume that derivability-reduction-condition with respect to φ holds. If $\vdash^{\mathbf{NFP}} \chi$, then $\vdash^{\mathbf{NFP}} \varphi$ clearly holds. Now ssume there is $X \subseteq \mathbb{X}$ and $Y \subseteq \mathbb{Y}$ such that:

- X is neat;
- $\bigcup_{x\in X} C_x \subseteq C_y$, $\bigcup_{x\in X} B_x \supseteq B_y$, and $\bigcup_{x\in X} H_x \supseteq H_y$ for all $y \in \mathbb{Y}$;
- there is at most one $y \in \mathbb{Y}$ such that $C_y \neq \mathbb{AGT}$;
- and if $\bigcup_{x\in X} B_x \cap \bigcup_{x\in X} H_x = \emptyset$ then $\vdash^{\mathbf{NFP}} \bigwedge_{x\in X} \psi_x \to \bigvee_{y\in Y} \psi_y$.

Consider whether $\bigcup_{x\in X} B_x \cap \bigcup_{x\in X} H_x = \emptyset$:

- Assume $\bigcup_{x\in X} B_x \cap \bigcup_{x\in X} H_x = \emptyset$. Then $\vdash^{\mathbf{NFP}} \bigwedge_{x\in X} \psi_x \to \bigvee_{y\in Y} \psi_y$. Similar to the proof in section ??, we have

$$\vdash^{\mathbf{NFP}} \bigwedge_{x\in X} \langle\!\langle C_x{:}B_x,H_x\rangle\!\rangle \mathsf{X}\psi_x \to \bigvee_{y\in Y} \langle\!\langle C_y{:}B_y,H_y\rangle\!\rangle \mathsf{X}\psi_y.$$

- Assume $\bigcup_{x\in X} B_x \cap \bigcup_{x\in X} H_x \neq \emptyset$. Similar to the proof in section ??,

$$\vdash^{\mathbf{NFP}} \bigwedge_{x\in X} \langle\!\langle C_x{:}B_x,H_x\rangle\!\rangle \mathsf{X}\psi_x \to \langle\!\langle \bigcup_{x\in X} C_x{:} \bigcup_{x\in X} B_x, \bigcup_{x\in X} H_x\rangle\!\rangle \mathsf{X} \bigwedge_{x\in X} \psi_x$$

Therefore,

$$\vdash^{\mathbf{NFP}} \bigwedge_{x\in X} \langle\!\langle C_x{:}B_x,H_x\rangle\!\rangle \mathsf{X}\psi_x \to \langle\!\langle \bigcup_{x\in X} C_x{:} \bigcup_{x\in X} B_x, \bigcup_{x\in X} H_x\rangle\!\rangle \mathsf{X}\top$$

Note $\bigcup_{x\in X} B_x \cap \bigcup_{x\in X} H_x \neq \emptyset$. By axiom ??,

$$\vdash^{\mathbf{NFP}} \neg\langle\!\langle \bigcup_{x\in X} C_x{:} \bigcup_{x\in X} B_x, \bigcup_{x\in X} H_x\rangle\!\rangle \mathsf{X}\top$$

Therefore,

$$\vdash^{\mathbf{NFP}} \bigwedge_{x\in X} \langle\!\langle C_x{:}B_x,H_x\rangle\!\rangle \mathsf{X}\psi_x \to \bigvee_{y\in Y} \langle\!\langle C_y{:}B_y,H_y\rangle\!\rangle \mathsf{X}\psi_y.$$

$\square$

D Proof of Theorem ??

Proof. The proof is the routine of "unfold" method. The right-to-left direction of the lemma is evident. We prove the left-to-right direction for the first item. The proof for the second item is analogous.

Let $(\mathsf{M}, s_0) \models \varphi$ with $\mathsf{M} = (\mathbb{S}, \mathbb{ACT}, \mathcal{R}, \preceq, \mathcal{V})$, and $s_0 \in \mathbb{S}$.

We are going to transform $\mathsf{M} = (\mathbb{S}, \mathbb{ACT}, \mathcal{R}, \preceq, \mathcal{V})$ into a new CGMP $\mathsf{M}' = (\mathbb{S}', \mathbb{ACT}, \mathcal{R}', \preceq', \mathcal{V}')$. The construction goes as follows:

- $\mathbb{S}' = PPath_{\mathsf{M}}$,
- for all $\lambda, \lambda' \in \mathbb{S}'$ and for all $\delta \in \mathbb{ACT}^{\mathbb{AGT}}$, $\lambda \mathcal{R}'_\delta \lambda'$ iff $\mathsf{last}(\lambda)\mathcal{R}_\delta \mathsf{last}(\lambda')$,
- for all $\pi \in \mathbb{S}'$, for all $i \in \mathbb{AGT}$, for all $\pi\pi_1\pi_2\ldots, \pi\pi'_1\pi'_2\ldots \in Comp_{\mathsf{M}',\pi}$, we have $\pi\pi_1\pi_2\ldots \preceq'_{i,\pi} \pi\pi'_1\pi'_2\ldots$ iff

$$\mathsf{last}(\pi)\mathsf{last}(\pi_1)\mathsf{last}(\pi_2)\ldots \preceq_{i,\pi} \mathsf{last}(\pi)\mathsf{last}(\pi'_1)\mathsf{last}(\pi'_2)\ldots$$

- for all $p \in \mathbb{P}$ and for all $\pi \in \mathbb{S}$, $p \in \mathcal{V}'(\pi)$ iff $p \in \mathcal{V}'(\mathsf{last}(\pi))$.

It is routine to verify that M' is a tree-like model. The function last mapping worlds in M' into worlds in M. Clearly, if M has stable preferences, then M' also has stable preferences

By induction on the structure of φ, it is straightforward to show that $(\mathsf{M}', \pi) \models \varphi$ iff $(\mathsf{M}, \mathsf{last}(\pi)) \models \varphi$ for all $\pi \in \mathbb{S}'$. $\square$

E Proof of Theorem ??

Proof. We prove the first statement of the theorem. Suppose $\varphi \in \mathcal{L}_{\mathsf{CA-ATL}}$ is satisfiable for the class $\mathbf{CP}^{\{\mathbf{SP}\}}$. Thus, by Lemma ??, there exists $\mathsf{M} = (\mathbb{S}, \mathbb{ACT}, \mathcal{R}, \preceq, \mathcal{V})$ and $s_0 \in \mathbb{S}$ such that $\mathsf{M} \in \mathbf{CP}^{\{\mathbf{TR},\mathbf{SP}\}}$ and $(\mathsf{M}, s_0) \models \varphi$. Define a new structure $\mathsf{M}' = (\mathbb{S}, \mathbb{ACT}, \mathcal{R}, \preceq, \mathcal{V}')$ such that for all $s \in \mathbb{S}$

$$\mathcal{V}'(s) = \mathcal{V}(s) \cup \{\heartsuit_a \mid \exists r \in \mathcal{R}^-(s) \exists \lambda \in Best_{a,s} \text{ s.t. } = \lambda(1)\} \cup \{\spadesuit_a \mid \exists r \in \mathcal{R}^-(s) \exists \lambda \in Worst_{a,s} \text{ s.t. } = \lambda(1)\}$$

Clearly, $\mathsf{M}' \in \mathbf{CP}^{\{\mathbf{SP},\mathbf{TR}\}}$ and the only difference between M' and M is in the valuation function that does not affect the relational properties of the structure. By induction on the structure of the formula, it is routine to prove that "$(\mathsf{M}, v) \models tr(\varphi)$ iff $(\mathsf{M}', v) \models \varphi$" for all $v \in W$. The other direction is proved in an analogous way.

The proof of the second statement is based on a straightforward adaptation of the above construction without requiring stable preferences. $\square$

F Proof of Proposition ??

Proof. Fix a CGFP $\mathsf{P} = (\mathsf{F}, \preceq)$. Assume strategy $\mathcal{F}_C$ of coalition C is strongly beneficial for group $B \subseteq \mathbb{AGT}$ at state s. Then $\mathcal{O}_{\mathsf{F}}(s, \mathcal{F}_C) \subseteq Best_{i,s}$ for every $i \in B$. Let $i \in B$, $\lambda \in \mathcal{O}_{\mathsf{F}}(s, \mathcal{F}_C)$ and $\lambda' \in Comp_{\mathsf{F},s} - \mathcal{O}_{\mathsf{F}}(s, \mathcal{F}_C)$. Then $\lambda \in Best_{i,s}$. Therefore, $\lambda' \not\succ_{i,s} \lambda$. Then $\mathcal{F}_C \in StrWB^C_{\mathsf{F}}(s, B)$.

The proof for harmful strategies is similar. $\square$